



Association
of International
Certified Professional
Accountants™

AICPA® & CIMA®

Strategic Case Study Examination

Analysis of Pre-seen Material

Nov 2024 and Feb 2025 CGMA Examinations

Clancy Peiris



战略级案例分析考试

预览材料分析

2024年11月/2025年2月考季

讲师：Clancy Peiris

Stages in analysis of SCS pre-seen case for Nov 2024 and Feb 2025 examinations

- Revision of the exam structure and assessment outcomes
 - Understanding the lithium-mining industry
 - Understanding Rotomyne's business model
 - Governance and Risk in Rotomyne
 - Financial analysis of Rotomyne
 - Review of industry articles
-

2024年11月2025年2月考季预览材料分析考试复习步骤：

- 回顾考试结构和评分结果
 - 理解锂矿开采行业
 - 理解罗托矿业商业模式
 - 罗托矿业的治理和风险管理
 - 罗托矿业的财务分析
 - 行业文章的回顾
-

The Examination Structure and Competences

考试结构和能力要求

This examination is structured as follows:

Section number	Time for section (minutes)	Number of tasks	Number of sub-task/s	% time to spend on each sub-task
1	60	1	2	(a) 50% (b) 50%
2	60	1	2	(a) 60% (b) 40%
3	60	1	2	(a) 40% (b) 60%

Each section (task) has a number of sub-tasks. An indication of how much of the time available for the section that you should allocate to planning and writing your answer is shown against each sub-task in the text of the question (and summarised in the table above).

This information will be available for you to access during the examination by clicking on the Pre-seen button.

战略案例分析考试 - 考生姓名

本次考试的结构如下：

部分编号	部分时间（分钟）	题目数量	小题数量	每个小题的时间百分比
1	60	1	2	(a) 50% (b) 50%
2	60	1	2	(a) 60% (b) 40%
3	60	1	2	(a) 40% (b) 60%

每个部分（题目）都有若干小题。在问题文本中（以及上表中总结），会显示你应该分配给计划和书写答案的时间比例。

在考试期间，你可以通过点击预览材料按钮访问此信息。

Reference Material

Pre-seen

A month later, you receive the following email:

From: Filiz Yildiz, Chief Finance Officer

To: Senior Finance Manager

Subject: Bribery allegations

Hello,

I have attached an extract from the minutes of this morning's Board meeting.

This is a potential disaster for us because we have several major clients in Triland. Furthermore, many of our clients are based in countries that have strict anti-corruption schemes.

A number of Board members have pointed out that there are some complicated ethical issues associated with interactions with potential buyers. For example, Robobryce arranges good accommodation and meals in excellent hotels for visiting buyers.

I need your advice on two matters:

- Firstly, recommend with reasons how we should manage the political risks associated with Ms Liqiong's allegations.

[sub-task (a) = 60%]

- Secondly, identify and evaluate the ethical issues associated with the level of hospitality we provide to visiting buyers.

[sub-task (b) = 40%]

Regards

Filiz

The extract referred to by Filiz can be viewed by clicking the Reference Material button above.

战略案例分析考试 - 考生姓名

一个月后，您收到以下电子邮件：

发件人：Filiz Yildiz，首席财务官

收件人：高级财务经理

主题：贿赂指控

你好，

我已附上今早董事会会议记录的摘录。

这对我们来说是一个潜在的灾难，因为我们在Triland有几个重要客户。此外，我们的许多客户都位于有严格反腐败方案的国家。

一些董事会成员指出，与潜在买家互动时存在一些复杂的道德问题。例如，Robobryce为来访的买家安排了优质的住宿和在高级酒店的餐饮。

我需要您在两方面提供建议：

- 首先，推荐并说明我们应如何管理与Ms Liqiong指控相关的政治风险。[小题 (a) = 60%]
- 其次，识别并评估我们为来访买家提供的接待水平所涉及的道德问题。[小题 (b) = 40%]

此致

Filiz

Filiz提到的摘录可以通过点击上方的参考材料按钮查看。

I need your advice on two matters:

- Firstly, recommend with reasons how we should manage the political risks associated with Ms Liqiong's allegations.

[sub-task (a) = 60%]

- Secondly, identify and evaluate the ethical issues associated with the level of hospitality we provide to visiting buyers.

[sub-task (b) = 40%]

我需要您在两方面提供建议：

- 首先，推荐并说明我们应如何管理与Ms Liqiong指控相关的政治风险。[小题 (a) = 60%]
- 其次，识别并评估我们为来访买家提供的接待水平所涉及的道德问题。[小题 (b) = 40%]

Board minute extract
Bribery accusation

Ewa Durska, Chief Executive Officer, informed the Board that Martin Anderson, one of Robobryce's senior sales managers, has been accused of attempted bribery. Martin was hosting a visit to Robobryce's research centre by Ms Shen Liqiong, a senior procurement official in the Trilandian Health Service. The Trilandian Health Service is keen to place a large order for a robot that can transport large loads of medical supplies and move patients through hospital corridors. Ms Liqiong was evaluating the new Heavybryce mobile robot for use in that role.

During the visit, Ms Liqiong mentioned that her daughter was an engineering student at Central City University in Tessland. Martin said that Ms Liqiong's daughter should apply for Robobryce's internship scheme for engineering students. Ms Liqiong was concerned that Martin was implying that her daughter's internship was conditional on Robobryce receiving a large order. She emailed her superiors that evening and was told to return to Triland immediately.

Martin denies offering Ms Liqiong any incentives to place an order but admits informing her that Robobryce has an excellent engineering internship programme.

The Trilandian Government has since contacted the Board to warn them that the bribery allegations would have to be investigated before Robobryce could be considered for any further contracts with Trilandian buyers.

董事会会议纪要摘录

贿赂指控

首席执行官Ewa Durska向董事会报告说，Robobryce公司的高级销售经理之一Martin Anderson被指控企图贿赂。Martin在接待来自特里兰迪亚卫生服务部门的高级采购官员Shen Liqiong女士参观Robobryce研究中心时，被指控贿赂。特里兰迪亚卫生服务部门有意订购一款能够在医院走廊中运输大量医疗用品和移动病人的机器人。Liqiong女士此次参观的目的是评估新款Heavybryce移动机器人在该角色中的使用情况。

在访问期间，Liqiong女士提到她的女儿是特斯兰中央城市大学的工程学学生。Martin表示Liqiong女士的女儿应该申请Robobryce的工程学生实习计划。Liqiong女士担心Martin暗示她女儿的实习机会与Robobryce获得大订单有关。当晚她通过邮件向上级报告了此事，并被告知立即返回特里兰。

Martin否认向Liqiong女士提供任何订单诱因，但承认告诉她Robobryce有一个优秀的工程实习项目。

特里兰政府随后联系了董事会，警告他们在贿赂指控被调查清楚之前，Robobryce不得被考虑与特里兰买家进行任何进一步的合同合作。

 Reference Material Pre-seen

Draft your response to Filiz's requests in the box below.

From: Senior Finance Manager
To: Filiz Yildiz, Chief Finance Officer
Subject: RE: Bribery allegations



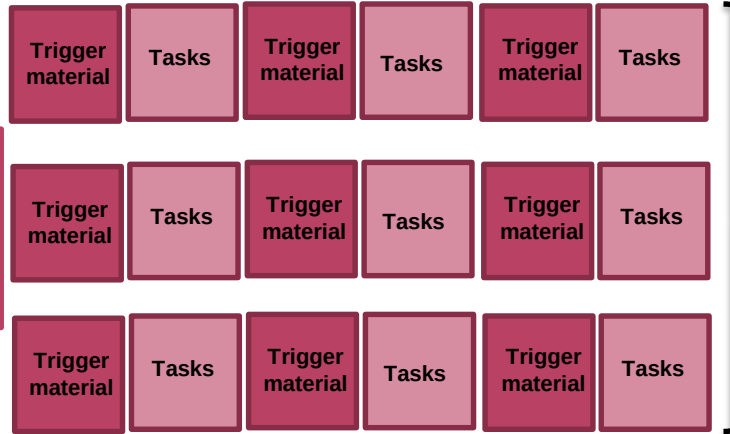
Exam structure

Pre-seen material

At least 6 weeks before exam week



Exam week

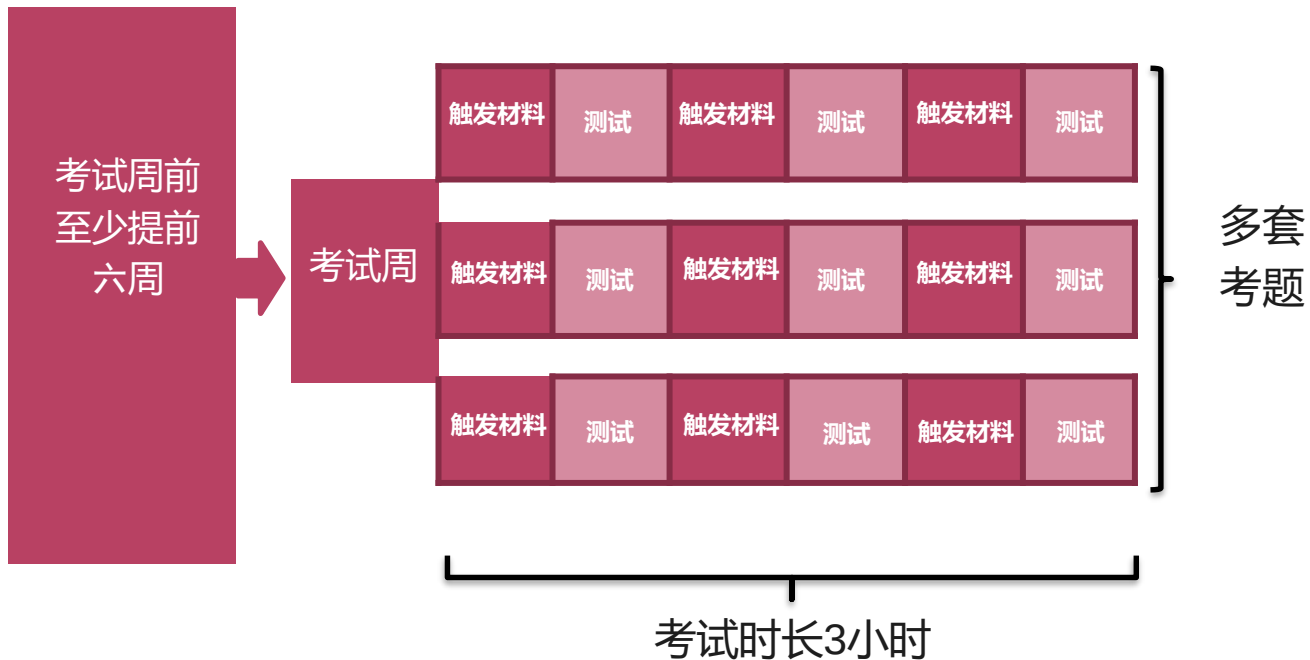


Multiple variants of exam

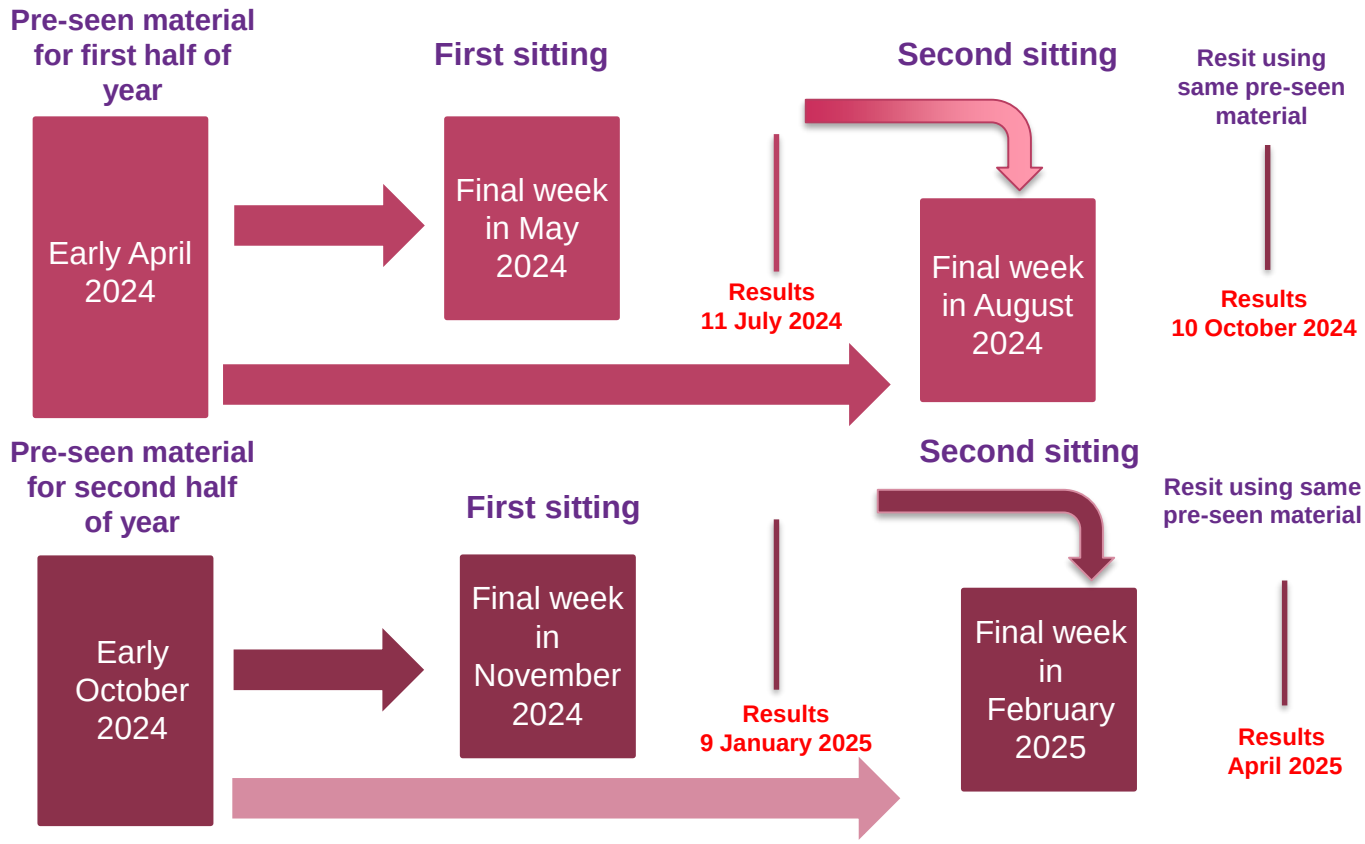
Up to 3 hours

考试结构

预览材料



Two exam cycles



两个考季

上半年预览材料

四月第二
周



第一次考试

五月最后
一周

第二次考试



补考, 预览材料不变

2024年7月11日
出成绩

八月最后
一周

2024年10月10日
出成绩

下半年预览材料

十月第二
周



第一次考试

十一月最
后一周

第二次考试



补考, 预览材料不变

2025年1月9日
出成绩

二月最后
一周

2025年4月中旬
出成绩

Main Skills to demonstrate and differences from the CGMA Management Case Study examination)

- **Clear and deep understanding of the significance of the pre-seen material**
 - Understand the business model of the firm
 - Understand the strengths and weaknesses of the firm and the opportunities it faces
 - Look for hints on topics that will be examined
-

CGMA 管理案例研究考试的主要技能展示与区别

- 明确并深刻理解预览材料的意义
 - 了解公司的业务模式
 - 了解公司的优缺点及其面临的机遇
 - 寻找将要考察的主题的提示
-

Main Skills to demonstrate and differences from the CGMA Management Case Study examination)

- **Understand unstructured questions**
 - Identify what the Financial Director (or the Board) needs to know (which may go beyond what they actually asked for)
 - Look for hints in the task upon what you need to consider and talk about
 - Cope with vague questions that can have several meanings
 - **Communication**
 - Write in clear concise paragraphs (one point in one paragraph)
 - Short sentences and simple business English
-

CGMA 管理案例研究考试的主要技能展示与区别

- **理解非结构化问题**

- 确定财务总监（或董事会）需要了解的内容（可能超出他们的实际要求）
- 寻找题目中的提示，了解您需要考虑和谈论的内容
- 应对可能有多种含义的模糊问题

- **沟通**

- 用简洁明了的段落写作（一个段落阐述一个要点）
 - 使用短句和简单的商务英语
-

Main Skills to demonstrate and differences from the CGMA Management Case Study examination)

- **Produce answers that demonstrate higher skills**
 - Keep theory to a minimum (80% application and only 20% repeating from a textbook)
 - Bring in points from outside the case study materials (not necessarily, real-world examples but your business common sense and general knowledge)
 - Be willing to discuss both sides of an issue
 - Be willing to make conclusions and to give advice and recommendations
-

CGMA 管理案例研究考试的主要技能展示与区别

- 答出能体现较高技能的答案
 - 尽量少用理论（80% 应用，20% 重复教科书内容）
 - 引入案例研究材料以外的观点（不一定是真实世界的例子，而是你的商业常识和一般知识）
 - 愿意讨论问题的正反两面
 - 愿意做出结论并提出意见和建议
-

Your job role (in Rotomyne)

Introduction

Rotomyne is a quoted company that mines lithium for sale to manufacturers, which then use the metal in the manufacture of various products, including rechargeable batteries.

You are a **senior manager** in Rotomyne's finance function. You report directly to the Board and **advise on special projects and strategic matters**.

Rotomyne's head office is located in Porrland, a developed country that has an active and well-regulated stock exchange. Porrland's currency is the P\$. Porrland requires companies to

你的角色 (在罗托矿业)

简介

罗托矿业(Rotomyne)是一家上市公司，其开采锂出售给制造商，然后制造商会使用这种金属制造各种产品，包括可再充电电池。

您是罗托矿业财务部的高级经理。您直接向董事会报告，并就特殊项目和战略事项提出建议。

罗托矿业的总部位于发达国家波兰德(Porrland)，该国拥有交易活跃且监管良好的证券交易所。波兰德的货币是 P\$。波兰德要求各公司根据国际财务报告准则 (IFRS) 编制财务报表。罗托矿业在六个国家拥有并经营锂矿。

Your job role (in Rotomyne)

Senior manager in finance function

Advise on special projects and strategic matters | Report directly to the Board.

A senior finance manager advising the organisation's leaders in the development of business strategy to create value is required to evaluate strategic options, paying due attention to the organisation's ecosystem. Risk management is a significant part of the overall responsibility, including cyber risks.

There is a duty to advise on issues relating to corporate governance. They are important both because of the reputational risks arising from poor governance and the threats to the well-being of internal and external stakeholders when boards are poorly structured and badly managed. As an important element of this, the senior finance manager is frequently involved in the evaluation of the control environment and the management of internal audit. The need to manage control is crucial given the increasing dangers arising from cyber risks.

The frequent requirement to support strategic decision-making may require the formulation of models and other methods of justifying and legitimating decisions. The senior finance manager is also involved in the raising of finance from suitable sources in order to ensure that new strategic projects are adequately resourced.

The business case for digitisation and the manner in which the entity creates partnerships for ensuring strategic success must be understood. As must the operation of capital markets in determining value.

The senior finance manager advises on matters that involve considerable judgement and that may have a significant effect on stakeholders and therefore must adhere to high standards of professionalism and ethics in the course of their duties.

Senior finance manager – simulating the job tasks

Within the Strategic Case Study Examination, five core activities will be assessed. These core activities represent the tasks most frequent, critical and important to the senior finance manager role.

The five core activities are:

- A. Develop business strategy
- B. Evaluate business ecosystem and business environment
- C. Recommend financing strategies
- D. Evaluate and mitigate risk
- E. Recommend and maintain a sound control environment

The core activities are linked to associated assessment outcomes expressed in terms of 'I Can' statements that speak directly to the skills and competencies that drive the employability of successful learners.

你在罗托矿业这家公司的角色

- 财务部门的高级经理，负责为特别项目和战略事务提供决策支持 — 直接向董事会汇报。



在战略案例研究考试中，将对五个核心活动进行评估。这些核心活动代表了高级财务经理职位最常见、最关键且最重要的任务。

这五个核心活动包括：

- A. 制定商业战略
- B. 评估商业生态系统及环境
- C. 制定融资策略
- D. 评估与减轻风险
- E. 推荐并维护良好的控制环境

这些核心活动与特定的评估成果相关联，这些成果以“我能”声明的形式表述，直接反映了成功学习者就业所需的技能和能力。

The Strategic Case Study Examination Blueprint

Core Activity	Assessment outcome 'I can....'	Weighting
A Develop business strategy	• Evaluate strategic options (digital and otherwise) • Recommend strategic decisions (digital and otherwise) • Evaluate potential acquisitions and divestment opportunities • Recommend responses to opportunities and threats arising from digital technologies	15-25%
B Evaluate business ecosystem and business environment	• Select and apply suitable strategic analytical tools • Conduct an analysis of stakeholder needs and recommend appropriate responses • Recommend appropriate responses to changes in the business ecosystem • Recommend KPIs that encourage sound strategic management • Recommend responses to economic, political and currency risks	15-25%

战略级案例考试考试蓝图

核心工作	考核的能力 '我能....'	权重
A 制定商业战略	• 评估战略选择（数字和其它） • 建议战略决策（数字和其它） • 评估收购和分拆的机会 • 针对数字技术带来的机遇和威胁建议应对方案	15-25%
B 评估商业生态系统 和商业环境	• 选择并应用合适的战略分析工具 • 分析利益相关方需求并建议恰当的应对方案 • 建议合适的方案，应对商业生态系统的变化 • 建议合适的KPI，促进健全的战略管理 • 建议应对经济、政治和汇率风险的方案	15-25%

The Strategic Case Study Examination Blueprint

Core Activity	Assessment outcome 'I can....'	Weighting
C Recommend financing strategies	• Recommend suitable sources of finance • Recommend dividend policy • Recommend and apply business valuation models	15-25%
D Evaluate and mitigate risk	• Evaluate risks and recommend responses and can maintain the corporate risk register • Identify ethical dilemmas and recommend suitable responses • Evaluate and mitigate cyber risks • Recommend internal controls	15-25%
E Recommend and maintain a sound control environment	• Apply internal audit resources • Recommend appropriate controls and evaluate the implications of compliance failures • Recommend responses to the threats arising from poor governance	15-25%

战略级案例考试考试蓝图

核心工作	考核的能力 '我能....	权重
C 建议融资策略	• 建议合适的融资方式 • 建议派息政策 • 建议并应用企业估值模型	15-25%
D 评估并缓减风险	• 评估风险，并建议应对措施，且能维护公司的风险管理列表 • 发现道德矛盾，并建议合适的应对方案 • 评估并缓减网络风险 • 建议内控措施	15-25%
E 建议并维护健全的控制环境	• 使用内审资源 • 建议合适的控制措施并评估合规问题的影响 • 针对因治理不当而发生的威胁建议应对措施	15-25%

Your job role (in Rotomyne)

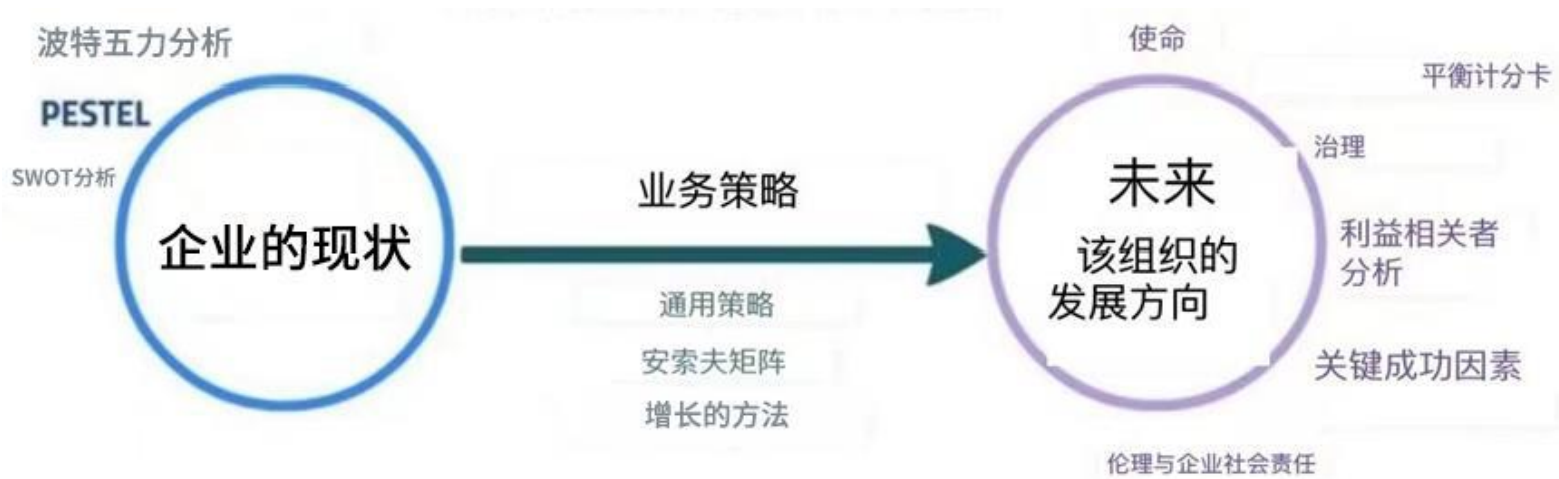
Senior manager in finance function

Advise on special projects and strategic matters | Report directly to the Board.



你的角色(在罗托矿业)

财务职能的高级经理为特殊项目和战略事务提供建议 | 直接向董事会汇报。



Mapping

- **Strategic Map** shows the link between (a) stages of strategy formulation, (b) Skills Workbook chapters, and (c) 'Rotomyne' Mock Exam tasks.
- **SCS Exam Blueprint Map** shows the link between (a) Assessment Outcomes, (b) key OT Exam syllabus areas, (c) Skills Workbook chapters, and (d) 'Rotomyne' Mock Exam tasks

**You will receive these maps
at the end of the programme.**

映射

- 战略图显示了 (a) 战略制定阶段、(b) 白皮书章节 和 (c) “Rotomyne” 模拟考试任务之间的联系。
- SCS考试蓝图图显示了 (a) 评估结果、(b) 关键OT考试大纲领域、(c) 白皮书章节 和 (d) “罗托矿业 ” 模拟考试任务之间的联系。

您将在课程结束时收到这些图表。

Understanding the Lithium Mining Industry

了解锂矿行业

Page 2

Introduction

What does this information mean to you?

- Mines lithium for sale to manufacturers who use lithium in various products (e.g., rechargeable batteries).
 - Owns and operates lithium mines in 6 countries.
 - Head office is in Porland - a developed country.
 - Quoted on Porland Stock Exchange
 - active market (= *efficient stock market*)
 - well-regulated (= *has codes for good corporate governance*)
 - Currency is P\$
 - Prepares financial statements in accordance with IFRS
-

第2页 介绍

这段信息对你
意味着什么？

- 公司开采锂矿，销售给制造商，这些制造商用锂生产各种产品（比如充电电池）。
 - 公司在6个国家拥有并运营锂矿。
 - 总部位于波兰德——一个发达国家。
 - 公司在波兰德证券交易所上市——这是一个活跃的市场（等于高效的股票市场）——监管完善（等于有良好的公司治理准则）。
 - 货币单位是P\$。
 - 公司按照国际财务报告准则（IFRS）编制财务报表。
-

Before we deep-dive into the industry ...

- Marks are NOT allocated for your knowledge or awareness of the real-world lithium mining industry.
 - Examiner wants you to support **your advice and recommendation** using industry information in the pre-seen.
 - If you provide examples from the real-world, you won't get marks for mere 'name-dropping' – it should support your point.
 - Focus on **Opportunities, Threats and Risks**
 - Useful models: PESTEL, Porter's 5 Forces, Porter's Diamond, SWOT etc.
-

在我们深入探讨该行业之前 ...

- 你对现实世界中的锂矿行业的了解或认识并不得分。
 - 考官希望您使用预览的行业信息来支持您的意见和建议。
 - 如果您提供了现实世界中的例子，仅仅 "引用 "这些例子是不会得分的--这些例子应能支持您的观点。
 - **关注机遇、威胁和风险**
 - 有用的模型： PESTEL、波特五力、波特钻石、SWOT 等。
-

Industry Life Cycle

- **Introduction phase:** small scale mining due to limited demand by niche uses such as glass and lubricants in the 19th century.
- **Growth phase:** kicked off in the 20th century with consumer electronics such as laptops, mobile phones, followed by EV manufacturing in the 21st century – new extraction technology (DLE) to meet the fast-growing demand.
- **Maturity phase:** entering to maturity phase, with fierce competition among large/established players who dominate the market. Focus has shifted towards innovation to improve extraction efficiency and reducing environmental impacts.

New applications for lithium, such as grid-scale energy storage, continue to emerge, keeping the industry dynamic.

What do you think the next phase?

行业生命周期

- 引入阶段：19世纪，由于玻璃和润滑剂等小众用途的需求有限，锂矿的开采规模较小。
- 增长阶段：20世纪，消费电子产品如笔记本电脑和手机的出现推动了锂矿需求的增长，21世纪电动汽车制造的兴起进一步加速了这一需求，推动了新提取技术（DLE）的开发。
- 成熟阶段：锂矿行业正进入成熟阶段，市场由大型成熟企业主导，竞争激烈。当前的重点转向通过创新提高提取效率，并减少对环境的影响。

锂的新应用不断涌现，例如电网级能源存储，推动了行业的持续发展和活力。你认为下一阶段会是什么？

Industry Life Cycle

- **Today:** As the world continues its shift towards renewable energy and electric vehicles, demand for lithium will increase (from 0.5 million MT in 2023 to 4 million MT by 2030).
- R&D is critical to the lithium mining industry's future. Innovation is focused on improving efficiency, reducing environmental impact, and ensuring sustainable supply to meet the rising global need for lithium.
- Governments play a crucial role in setting regulations related to environmental standards, labour practices and operational procedures .

Energy-intensive nature of lithium mining and processing, significant water use in brine extraction ... companies must implement sustainable practices to minimize their environmental footprint.

行业生命周期

- 当今，随着全球向可再生能源和电动汽车的转型，对锂的需求将大幅增加（从2023年的50万吨增至2030年的400万吨）。
- 研发对锂矿行业的未来至关重要，创新的重点在于提高效率、减少环境影响，并确保可持续供应以满足全球对锂日益增长的需求。
- 各国政府在制定环境标准、劳动规范和操作流程等相关法规方面发挥着关键作用。

锂矿开采和加工能耗巨大，卤水提取过程中消耗大量水资源.....企业必须实施可持续的措施，尽量减少对环境的影响。

Key Customers

- **EV manufacturers** Tesla, Volkswagen and BYD are ramping up production as governments push for zero-emission targets and greener transport incentives.
 - **Battery manufactures:** CATL, LG Chem and Panasonic are the real powerhouses behind the scenes, producing the lithium-ion batteries found in EVs, laptops, smartphones, etc.
 - **Energy storage companies:** NextEra Energy and Tesla Energy are building huge lithium-ion battery systems to store solar and wind power, ensuring grid stability when the sun isn't shining, or the wind isn't blowing.
-

重要客户

- 电动汽车制造商特斯拉、大众和比亚迪正在加速生产，响应各国政府推动的零排放目标和绿色交通激励措施。
 - 电池制造商：宁德时代、LG化学和松下是幕后真正的核心力量，生产的锂离子电池广泛应用于电动汽车、笔记本电脑和智能手机等设备中。
 - 能源储存公司：NextEra能源和特斯拉能源正在建设大型锂离子电池系统，用于存储太阳能和风能，确保在没有阳光或风力的情况下电网依然保持稳定。
-

Key Customers

- **Consumer electronics producers:** Apple, Samsung and Sony drive significant lithium demand for smartphones, laptops, tablets, etc. Consumers want lightweight long-lasting and fast-charging devices.
 - **Industrial users:** Lithium hydroxide is a key component in high-temperature lubricants used in automotive, aerospace and heavy machinery, keeping equipment running smoothly.
-

重要客户

- 消费电子制造商：苹果、三星和索尼对锂的需求巨大，主要用于智能手机、笔记本电脑、平板电脑等产品。消费者希望设备具备轻便、续航持久和快速充电的特点。
 - 工业用户：氢氧化锂是高温润滑剂的关键成分，广泛应用于汽车、航空航天和重型机械领域，确保设备的平稳运行。
-

Geographical locations

- **Lithium Triangle:** (Chile, Argentina, Bolivia) - Home to about 50-60% of the world's lithium reserves, primarily extracted from brine in arid environments, posing water scarcity challenges.
 - **Australia:** The world's largest hard rock lithium producer, with strong mining infrastructure but environmental concerns from extraction processes.
 - **China:** Dominates lithium processing, refining over 60% of global capacity, despite holding fewer lithium resources.
 - **United States:** Modest reserves but increasing domestic production to reduce reliance on foreign sources, driven by EV growth.
-

Emerging regions:
Europe and Africa are gaining attention for their lithium potential, with countries like Portugal, Germany, Zimbabwe and the DRC exploring resources to diversify global supply.

地理位置

- 锂三角：（智利、阿根廷、玻利维亚）——该地区拥有全球约50-60%的锂储量，主要通过干旱环境中提取卤水锂，这带来了水资源短缺的挑战。
- 澳大利亚：全球最大的硬岩锂生产国，采矿基础设施完备，但提取过程引发了环境问题。
- 中国：尽管锂资源较少，但主导全球锂加工，精炼能力占全球总产能的60%以上。
- 美国：锂储量较为有限，但正在增加国内生产，以减少对外国资源的依赖，这一趋势受电动汽车市场增长的推动。

新兴地区：欧洲和非洲的锂资源潜力备受关注，葡萄牙、德国、津巴布韦和刚果民主共和国等国家正在开发资源，以实现全球供应的多元化。

Marketing Mix: Products

- Rotomyne offers more than just lithium extraction. It delivers high-purity battery-grade lithium hydroxide and carbonate essential for EVs and energy storage systems.
 - By focusing on sustainable extraction methods, like Direct Lithium Extraction (DLE), Rotomyne aligns with industry leaders prioritising eco-friendly practices.
 - This focus on purity and sustainability ensures Rotomyne stands out as a premium supplier in a competitive market.
-

营销组合: 产品

- 罗托矿业不仅提供锂矿开采，还供应高纯度电池级氢氧化锂和碳酸锂，这些对于电动汽车和能源储存系统至关重要。
 - 通过专注于可持续的开采方法，如直接锂提取（DLE），罗托矿业与优先采用环保措施的行业领袖保持一致。
 - 这种对纯度和可持续性的关注，使罗托矿业在竞争激烈的市场中脱颖而出，成为优质供应商。
-

Marketing Mix: Price

- The price of lithium carbonate rose by over 400% in 2021 due to surging demand, showcasing the volatility of the market.
- Multi-year contracts are gaining popularity as they offer pricing stability, helping mitigate risks in a volatile market, especially with growing demand for EVs.
- Rotomyne's pricing structure reflects the high quality of its lithium products, accounting for factors like extraction complexity, purity levels and market demand.

With the volatility of lithium prices, offering stable, long-term contracts allows Rotomyne to position itself as a reliable partner while ensuring fair pricing models that emphasise both sustainability and return on investment for customers.

营销组合：价格

- 由于需求激增，碳酸锂的价格在2021年上涨了400%以上，突显了市场的波动性。
- 多年期合同越来越受欢迎，因为它们提供价格稳定性，有助于在波动的市场中降低风险，尤其是在电动汽车需求不断增长的情况下。
- 罗托矿业的定价结构反映了其锂产品的高质量，并综合考虑了提取的复杂性、纯度水平和市场需求等因素。

面对锂价的波动，提供稳定的长期合同使罗托矿业能够将自己定位为一个可靠的合作伙伴，同时确保定价模型公平，兼顾可持续性和客户的投资回报。

Marketing Mix: Place

- Proximity to lithium-rich regions like Chile or Australia can reduce lead times and ensure a reliable supply chain.
 - Australia, the world's largest hard rock lithium producer, accounted for about 48% of global lithium production in 2020 , giving it a significant strategic advantage.
 - Some producers are moving processing facilities closer to EV hubs in Europe and North America to better serve their clients, reducing logistics challenges and taking advantage of regional incentives.
-

Rotomyne distributes its products to over 90 countries , leveraging its global reach to supply lithium compounds to various industries.

This strategic placement ensures efficiency and cost-effectiveness in its distribution network , positioning Rotomyne as a reliable global supplier.

营销组合: 渠道

- 靠近智利或澳大利亚等锂资源丰富的地区可以缩短交货周期，确保供应链的可靠性。
- 澳大利亚是全球最大的硬岩锂生产国，2020年约占全球锂产量的48%，这为其提供了显著的战略优势。
- 一些生产商正在将加工设施迁至欧洲和北美的电动汽车枢纽附近，以更好地服务客户，减少物流挑战，并利用区域性激励措施。

罗托矿业将其产品分销至全球超过90个国家，利用其全球布局为多个行业提供锂化合物。

这种战略布局确保了其分销网络的高效性和成本效益，确立了罗托矿业作为可靠全球供应商的地位。

Marketing Mix: Promotion

- In the lithium industry, companies don't just sell raw materials; they market themselves as key players in the green energy and electric vehicle revolutions.
- A strong brand highlights a commitment to quality, sustainability and cutting-edge technology, which is crucial for securing long-term partnerships with major customers.
- **Albemarle** positions itself as a leader in sustainable lithium extraction, appealing to eco-conscious clients like Tesla . This focus on sustainability enhances their brand image and makes them stand out in a crowded market.

Ganfeng Lithium has built its reputation around innovation and a fully integrated supply chain, offering both lithium mining and battery production.

This approach makes Ganfeng a go-to partner for automakers looking for a reliable, all-in-one solution.

营销组合: 推广

- 在锂行业，企业不仅仅销售原材料；它们还将自己营销为绿色能源和电动汽车革命中的重要参与者。
- 强大的品牌能够凸显对质量、可持续性和前沿技术的承诺，这对于与主要客户建立长期合作关系至关重要。
- 雅宝公司将自己定位为可持续锂开采的领导者，吸引了像特斯拉这样注重环保的客户。这种对可持续性的专注提升了其品牌形象，使其在竞争激烈的市场中脱颖而出。

赣锋锂业凭借创新和全产业链整合建立了其声誉，提供锂矿开采与电池生产的一体化服务。

这一策略使赣锋锂业成为汽车制造商寻求可靠的一站式解决方案的首选合作伙伴。

Key Suppliers

- **Mining equipment suppliers:** Caterpillar and Komatsu provide the heavy machinery needed for extraction, including drilling, blasting and transporting ore. Supplier power is moderate - multiple global players exist (complexity and reliability of equipment in remote regions give these suppliers leverage).
- **Chemical reagent suppliers:** BASF and Dow Chemical supply the chemicals required for purifying lithium. Supplier power is moderate to high - specialised reagents that ensure high purity standards for lithium-ion batteries.

Technology partners:

Innovators like Lilac Solutions offer new extraction technologies like DLE. Supplier power is moderate to high - depending on the effectiveness of these breakthrough technologies in improving efficiency and sustainability.

重要供应商

- 采矿设备供应商：卡特彼勒和小松提供锂开采所需的重型机械设备，包括钻探、爆破和矿石运输。供应商的议价能力为中等——全球有多个供应商存在，然而设备在偏远地区的复杂性和可靠性为这些供应商提供了优势。
- 化学试剂供应商：巴斯夫和陶氏化学提供用于锂提纯的化学试剂。供应商的议价能力中等到较高——这些专业试剂确保了锂离子电池的高纯度标准。

技术合作伙伴：像 Lilac Solutions 这样的创新公司提供了直接锂提取（DLE）等新型提取技术。供应商的议价能力中等至较高——这取决于这些突破性技术在提升效率和可持续性方面的有效性。

Key Suppliers

- **Water management suppliers:** In water-scarce regions like Chile, suppliers like Veolia provide essential water treatment and filtration systems for extraction and refining. Supplier power is high - environmental pressures and the need for efficient water management.
- **Engineering and construction firms:** Fluor design and build mining infrastructure to meet growing demand. Supplier power is moderate - many firms can handle construction, but large-scale projects require expertise in managing environmental regulations.

Renewable energy providers:

NextEra Energy provide renewable power solutions. Supplier power is high - limited number of large providers to help lithium producers lower their carbon footprint.

重要供应商

- 水管理供应商：在像智利这样水资源稀缺的地区，Veolia等供应商提供用于开采和提炼的关键水处理和过滤系统。供应商的议价能力较高——主要由于环境压力和高效水管理的需求。
- 工程和建筑公司：福陆公司设计并建设采矿基础设施，以应对日益增长的需求。供应商的议价能力中等——虽然许多公司可以从事建设工作，但大规模项目需要在管理环境法规方面的专业知识。

可再生能源供应商：
NextEra能源公司提供
可再生能源解决方案。
供应商的议价能力较
高——因为能够帮助
锂生产商减少碳足迹
的大型供应商数量有
限。

Transport and distribution of lithium

- A well-coordinated transport and distribution network is critical to move raw materials and refined lithium products from mining sites to processing plants, users of lithium across the globe.
 - Methods: freight shipping. Trucking, rail freight, inter-modal transportation, etc.
 - Cost: transporting large quantities of lithium across long distances from remote mining sites.
 - Damage during transit: poor handling or packaging during transport can damage high-value, battery-grade products.
 - Regulatory compliance: varying regulations around the transport of lithium and hazardous materials.
-

锂的运输与分销

- 一个协调良好的运输和分销网络对于将原材料和精炼后的锂产品从矿区运送到加工厂以及全球锂用户手中至关重要。
 - 运输方式：货运航运、卡车运输、铁路货运、联运等。成本：从偏远的矿区长距离运输大量锂产品。
 - 运输过程中的损坏：运输中的不当处理或包装可能损坏高价值的电池级产品。
 - 法规合规：各地关于锂及危险品运输的法规不同。
-

Competitors: Albermarle Corporation

- Founded in 1994, and a leading producer of lithium and a significant player in the chemicals industry, with operations in Australia, Chile, and the United States - specialises in both brine and hard rock lithium extraction.
 - Vertically integrated operations (e.g., processing plants in China and the U.S) allow it to consistently supply high-purity lithium carbonate and lithium hydroxide for use in EVs, consumer electronics, and energy storage systems.
 - Committed to sustainability (key differentiator): pioneering DLE (investment in R&D), a method designed to reduce water usage and minimise environmental impact.
-

竞争对手：雅宝公司

- 雅宝公司成立于1994年，是领先的锂生产商，也是化工行业的重要参与者，业务覆盖澳大利亚、智利和美国，专注于卤水和硬岩锂的开采。
 - 其垂直整合的运营模式（例如在中国和美国的加工厂）使其能够持续供应高纯度的碳酸锂和氢氧化锂，广泛应用于电动汽车、消费电子产品和能源存储系统。
 - 雅宝公司致力于可持续发展（其关键差异点）：率先开发直接锂提取（DLE）技术，并投资研发，这种方法旨在减少水资源的使用并降低环境影响。
-

Competitors: Albermarle Corporation

- Strategy is built on three pillars: vertical integration, sustainability and innovation. By controlling both mining and processing operations, it ensures a consistent supply of high-quality, battery-grade lithium to meet the needs of the EV and renewable energy sectors.
 - Maintains long-term supply agreements with major automakers and battery producers, offering stability in a volatile market.
 - Markets its brand by focusing on sustainability, innovation and quality - emphasising its leadership in green lithium production and promoting its role in the global energy transition.
-

Mission:

“We power the potential of people to lead fulfilling lives through the sustainable advancement of the essential elements of modern living.”

竞争对手：雅宝公司

- 战略建立在三个支柱之上：垂直整合、可持续发展和创新。通过控制采矿和加工环节，确保为电动汽车和可再生能源行业提供持续的高质量电池级锂供应。
- 公司与主要汽车制造商和电池生产商保持长期供应协议，确保在波动的市场中提供稳定性。
- 公司通过专注于可持续性、创新和质量来推广品牌，强调其在绿色锂生产中的领导地位，并推动其在全球能源转型中的重要角色。

使命：

“我们通过可持续发展现代生活的核心要素，赋能人们过上充实的生活。”

Competitors: Albermarle Corporation

- In late 2024 , Albermarle, the largest lithium producer, saw its stock rise by 13% after Contemporary Amperex Technology (CATL), a major Chinese battery manufacturer, announced plans to reduce lithium production at its Yichun City mine.
 - Despite these gains, Albermarle's stock has fallen nearly 40% in 2024 due to falling prices and reported losses.
-

竞争对手：雅宝公司

- 2024年末，全球最大的锂生产商雅宝公司在中国主要电池制造商宁德时代宣布计划减少宜春矿区的锂产量后，股价上涨了13%。
 - 尽管有这些涨幅，雅宝的股价在2024年仍下跌了近40%，主要由于价格下跌和报告的亏损。
-

Competitors: Albermarle Corporation

Albemarle Corporation and Subsidiaries

CONSOLIDATED STATEMENTS OF COMPREHENSIVE INCOME

(In Thousands)

Year Ended December 31

	2023	2022
Net income	\$ 1,670,543	\$ 2,815,131
Other comprehensive income (loss), net of tax:		
Foreign currency translation and other	26,403	(171,295)
Net investment hedge	—	—
Cash flow hedge	5,851	(4,399)
Interest rate swap	—	7,399
Total other comprehensive income (loss), net of tax	32,254	(168,295)
Comprehensive income	1,702,797	2,646,836
Comprehensive income attributable to noncontrolling interests	(97,185)	(125,232)
Comprehensive income attributable to Albemarle Corporation	\$ 1,605,612	\$ 2,521,604

竞争对手：雅宝公司

阿尔贝马尔公司及其子公司
综合全面收益表

(在汤姆桑德斯?)

截至二零二三年十二月三十一日止年度

	2023	2022
净收入	\$ 1,670,543	\$ 2,815,131
除税后其他全面收益 (亏损) :		
外币折算及其他	26,403	(171,295)
净投资对冲	—	—
现金流对冲	5,851	(4,399)
利率互换	—	7,399
除税后其他全面收益 (亏损) 总额	32,254	(168,295)
综合收入	1,702,797	2,646,836
归属于非控制性权益的全面收益	(97,185)	(125,232)
应归属于阿尔贝马尔公司的综合收入	\$ 1,605,612	\$ 2,521,604

Competitors: Albermarle Corporation

- Significant decrease in net income, from \$2.82 billion in 2022 to \$1.67 billion in 2023 (due to fluctuations in market demand and external conditions).
 - Comprehensive income in 2023 was \$1.70 billion, down from \$2.64 billion in 2022 (due to challenging operational environment), but its financial management and cost control measures continue to demonstrate resilience.
 - Total assets increased from \$15.5 billion in 2022 to \$18.2 billion in 2023 (due to significant capital investments in PPE).
 - Current liabilities increased to \$3.56 billion in 2023 from \$2.74 billion in 2022 (due to higher short-term debt obligations).
-

Long-term debt: moderate rise, from \$3.21 billion to \$3.54 billion (due to Albemarle is strategically managing its debt to finance expansion while balancing long-term liabilities).

Shareholders' equity: rose from \$7.98 billion to \$9.41 billion, driven by retained earnings and strategic growth investments.

竞争对手：雅宝公司

- 净收入大幅下降，从2022年的28.2亿美元降至2023年的16.7亿美元（由于市场需求波动和外部环境的变化）。
- 2023年综合收益为17亿美元，低于2022年的26.4亿美元（由于运营环境挑战），但其财务管理和成本控制措施依然表现出韧性。
- 总资产从2022年的155亿美元增加至2023年的182亿美元（得益于对PPE的重大资本投资）。
- 流动负债从2022年的27.4亿美元增加至2023年的35.6亿美元（由于短期债务义务的增加）。

长期债务：

从32.1亿美元适度上升至35.4亿美元（雅宝公司通过战略性管理债务，融资扩展的同时平衡长期负债）。

股东权益：

从79.8亿美元增加至94.1亿美元，主要由留存收益和战略增长投资推动。

Other Competitors

SQM or Sociedad Química y Minera (revenue: \$10.6bn)

Operates major projects like Greenbushes (Australia) and Salar de Atacama (Chile) , supporting EV and battery markets.

Tianqi Lithium (revenue: \$6.7bn)

Based in China, and its major projects include Greenbushes (Australia) and the Kwinana Lithium Hydroxide Plant.

Ganfeng Lithium (revenue: \$6.3bn)

A Chinese company with a diversified product portfolio, including lithium metal , lithium hydroxide and lithium carbonate . Major projects include Cauchari-Olaroz (Argentina) and Mt Marion (Australia) .

其他竞争对手

SQM或智利化工矿业公司（收入：106亿美元）运营着Greenbushes（澳大利亚）和阿塔卡马盐沼（智利）等主要项目，支持电动汽车和电池市场。

天齐锂业（收入：67亿美元），总部位于中国，主要项目包括Greenbushes（澳大利亚）和Kwinana氢氧化锂厂。

赣锋锂业（收入：63亿美元），是一家中国公司，拥有多样化的产品组合，包括锂金属、氢氧化锂和碳酸锂。其主要项目包括Cauchari-Olaroz（阿根廷）和Mt Marion（澳大利亚）。

Interesting stories

- In 2023, UK fire services responded to nearly three fires a day caused by lithium-ion batteries, marking a 46% increase from the previous year. These incidents, linked to electric vehicles, bikes and scooters, have raised safety concerns.
 - Stanford University has developed a new lithium extraction method called redox-couple electrodialysis, which is 40% cheaper and more environmentally friendly than traditional methods. It reduces the cost of producing battery-grade lithium carbonate to \$4,500 per tonne (\$9,100 per tonne using conventional brine extraction). By eliminating the need for large evaporation ponds and drastically lowering electricity usage, this method offers a cleaner and more efficient alternative.
-

有趣的故事

- 2023年，英国消防部门每天平均应对近三起由锂离子电池引发的火灾，比前一年增加了46%。这些事故与电动汽车、电动自行车和电动滑板车有关，已引发了安全担忧。
 - 斯坦福大学开发了一种名为氧化还原对电渗析的新锂提取方法，这种方法比传统方法便宜40%，且更加环保。它将电池级碳酸锂的生产成本降低至每吨4,500美元（相比之下，传统卤水提取的成本为每吨9,100美元）。通过消除大规模蒸发池的需求并大幅降低电力消耗，这种方法提供了一种更清洁、更高效的替代方案。
-

Interesting stories

- Study by Infyos revealed that up to 75% of global lithium-ion battery suppliers are linked to severe human rights abuses , particularly in Africa.
 - The increasing demand for EVs has led to a surge in battery demand, which reached over 750 GWh in 2023, up 40% from the previous year.
 - Investments in mining and refining have ensured that global supply currently exceeds demand. This comes at the same time as the emergence of potential alternatives to lithium, such as sodium-ion batteries.
-

有趣的故事

- Infyos的一项研究揭示，多达75%的全球锂离子电池供应商与严重的人权侵犯有关，尤其是在非洲的部分地区。
 - 电动汽车需求的增长导致电池需求激增，2023年电池需求超过750 GWh，较前一年增长了40%。
 - 采矿和精炼方面的投资确保了全球供应目前超过需求。与此同时，锂的潜在替代品（如钠离子电池）也开始出现。
-

Interesting stories

- The lithium-ion battery materials market is projected to grow from \$41.9 billion in 2024 to \$120.9 billion by 2029 , at a compound annual growth rate (CAGR) of 23.6%.
 - In July 2024, The European Union signed a historic agreement with Serbia, granting Rio Tinto permission to extract lithium from Serbia's Jadar Valley. Serbian President Aleksandar Vučić continues to support the project, citing job creation and economic benefits, while Rio Tinto claims the mine could supply 60,000 tonnes of lithium annually, addressing nearly one-fifth of Europe's lithium demand for electric vehicle batteries. Despite these potential benefits, mass protests against Rio Tinto's proposed lithium mine in the Jadar Valley, with environmental concerns.
-

有趣的故事

- 锂离子电池材料市场预计将从2024年的419亿美元增长至2029年的1209亿美元，年均复合增长率为23.6%。2024年7月，欧盟与塞尔维亚签署了一项历史性协议，授予力拓公司在塞尔维亚雅达尔山谷开采锂的许可。
 - 塞尔维亚总统亚历山大·武契奇继续支持这一项目，称其将创造就业机会并带来经济利益，而力拓公司声称该矿每年可供应6万吨锂，几乎满足欧洲电动汽车电池需求的五分之一。尽管有这些潜在好处，因环保问题，力拓公司在雅达尔山谷的锂矿项目遭遇了大规模抗议活动。
-

Interesting stories

- Ganfeng Lithium Group has agreed to acquire Leo Lithium's remaining 40% stake in the Goulamina lithium mine in Mali for \$342.7 million. This acquisition highlights both the economic potential of the project and the risks posed by the 2023 Malian Mining Code and the security situation in Mali.
 - According to the 2024 edition of PwC's 'Mine' report, the challenges and innovations shaping the global mining industry . The industry's revenue has been hit by falling commodity prices and rising costs, yet miners continue to invest in growth, particularly in critical minerals like lithium and copper. Mergers and acquisitions remain a key strategy for miners to gain technical capabilities and collaborate across industries.
-

有趣的故事

- 赣锋锂业集团已同意以3.427亿美元收购Leo Lithium在马里Goulamina锂矿的剩余40%股份。这笔收购不仅展示了该项目的经济潜力，同时也反映了2023年马里采矿法和当地安全局势带来的风险。
 - 根据普华永道2024年《矿业》报告，全球矿业行业面临的挑战和创新正在塑造行业发展格局。尽管大宗商品价格下跌和成本上升影响了行业收入，矿企仍在继续投资增长，尤其是在锂和铜等关键矿产方面。并购依然是矿企获取技术能力和跨行业合作的关键战略。
-

Lithium mining

- Lithium is used for variety of products (e.g., batteries, lubricants, pharmaceuticals etc.)
 - (Risk/Threat) Derived demand from major buyers from variety of industries.
 - (Opportunity) Vertical integration (for growth and control)
 - Traditional battery can store 30 watt-hours in 1l space, but a lithium battery can store 250 watt-hours in the same space.
 - (Opportunity) Efficiency and effectiveness of battery technology.
-

第3页

锂矿开采

- 锂被用于多种产品（如电池、润滑剂、药品等）。
 - （风险/威胁） 来自不同行业主要买家的衍生需求。
 - （机会） 垂直整合（促进增长与控制）。
 - 传统电池在1升空间内可存储30瓦时的电量，而锂电池在相同空间内可存储250瓦时的电量。（机会） 电池技术的高效性与效能。
-

Lithium mining

- Global demand for lithium carbonate was 500,000 MT in 2021. Annual demand has been predicted to increase to 4 million MT by 2030.
 - (Opportunity) Governments have started prohibiting or restricting internal combustion engines (and plan to move everyone for EVs in the next 10-15 years).
 - (Threat) More demand = Increased competition
 - (Threat) Governments tighten laws due to environmental impact
 - (Strength) Rotomyne is a key player established business
-

第3页

锂矿开采

- 2021年全球对碳酸锂的需求为50万吨，预计到2030年将增加至400万吨。
 - （机会） 各国政府已开始禁止或限制内燃机，并计划在未来10至15年内全面转向电动汽车。
 - （威胁） 需求增加 = 竞争加剧
 - （威胁） 政府因环境影响而收紧法律法规。（优势） 罗托矿业是行业中的重要参与者，且业务已稳固。
-

Lithium mining

- Lithium deposits can be found in many locations around the world.
 - (Threat) Limited locations
 - (Threat) Governments restrict access rights
 - It is also present in seawater, although extraction from seawater requires too much energy to be commercially viable.
 - (Opportunity) Invest in R&D and find ways to commercially viable
-

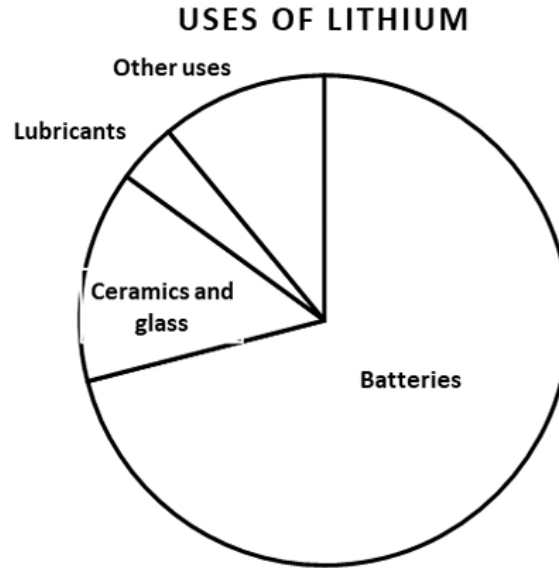
第3页

锂矿开采

- 锂矿床遍布世界多个地区。
 - （威胁） 位置有限
 - （威胁） 政府限制开采权。
 - 锂也存在于海水中，但从海水中提取锂的能耗过高，尚不具备商业可行性。
 - （机会） 投资研发，找到使其具备商业可行性的方法。
-

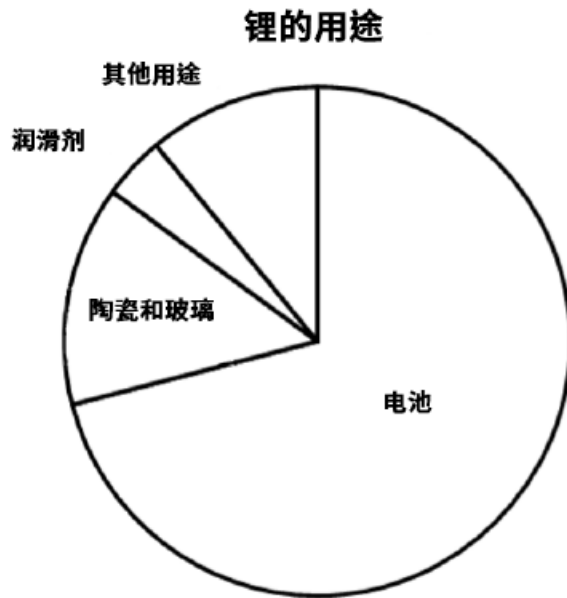
Page 3

Lithium mining



- (Opportunity) Diversify into battery manufacturing
 - (Opportunity) Different quality grades for different users
-

第3页 锂矿开采



- （机会） 多元化发展，进入电池制造领域
 - （机会） 为不同用户提供不同质量等级的产品
-

Lithium production methods

- **Hard rock mines** - lithium can constitute up to 8% of the content of spodumene.
 - Open pit mining has significant adverse environmental impacts (e.g., soil erosion, water pollution etc.)
 - (Opportunity) High barriers-to-entry = reduced threat of new entrants
 - (Risk) Environmental issue, government regulations, legal issues, ethical issues etc.
 - (Opportunity/Threat) CSR
-

第 4 – 7 页

锂的生产方法

- 硬岩矿 - 锂在锂辉石中的含量可高达8%。
 - 露天采矿对环境有显著的不利影响（如土壤侵蚀、水污染等
 - （机会） 高进入壁垒 = 减少新进入者的威胁
 - （风险） 环境问题、政府法规、法律问题、伦理问题等。
 - （机会/威胁） 企业社会责任（CSR）
-

Lithium production methods

- **Traditional brine mines** – in evaporation ponds (to 25 square kilometers), brine is exposed to heat and sunlight for 12 to 18 months.
 - Cheaper, but cause environmental damage (e.g., can pollute local water supplies). Each MT of lithium requires 500,000 litres of water to process, that would otherwise be used for human consumption or agriculture.
 - (Risk) Environmental issue, government regulations, legal issues, ethical issues etc.
 - (Opportunity/Threat) CSR
-

第4 – 7 页

锂的生产方法

- 传统的卤水矿 – 在蒸发池中（面积可达25平方公里），卤水暴露于热量和阳光下，需12到18个月进行处理。
 - 虽然成本较低，但会对环境造成破坏（如污染当地水源）。每生产一吨锂需要50万升水，这些水本可以用于人类饮用或农业灌溉。
 - （风险） 环境问题、政府法规、法律问题、伦理问题等。
 - （机会/威胁） 企业社会责任（CSR）
-

Lithium production methods

- **Direct Lithium Extraction (DLE)** – more expensive, but it avoids the environmental damage associated with storing brine in evaporation ponds. It also uses solvents to extract the lithium from the brine, so the extraction plants use much less fresh water.
 - (Opportunity) Invest in R&D to make this method commercially viable.
 - (Opportunity) Promotions and marketing (requires least water, area etc. and no emissions)
-

第4 – 7 页

锂的生产方法

- **直接锂提取（DLE）**——虽然成本较高，但它避免了将卤水储存在蒸发池中所带来的环境破坏。该技术通过溶剂从卤水中提取锂，显著减少了提取工厂对淡水的使用。
 - （机会） 投资研发，使这种方法具备商业可行性。
 - （机会） 推广与营销（用水最少、占地小且无排放）。
-

Exercise 1

Describe the elements of the ecosystem and business environment within which Lithium mining companies operate.

SCS Exam Blueprint: B1

“I can select and apply suitable strategic analytical tools.”

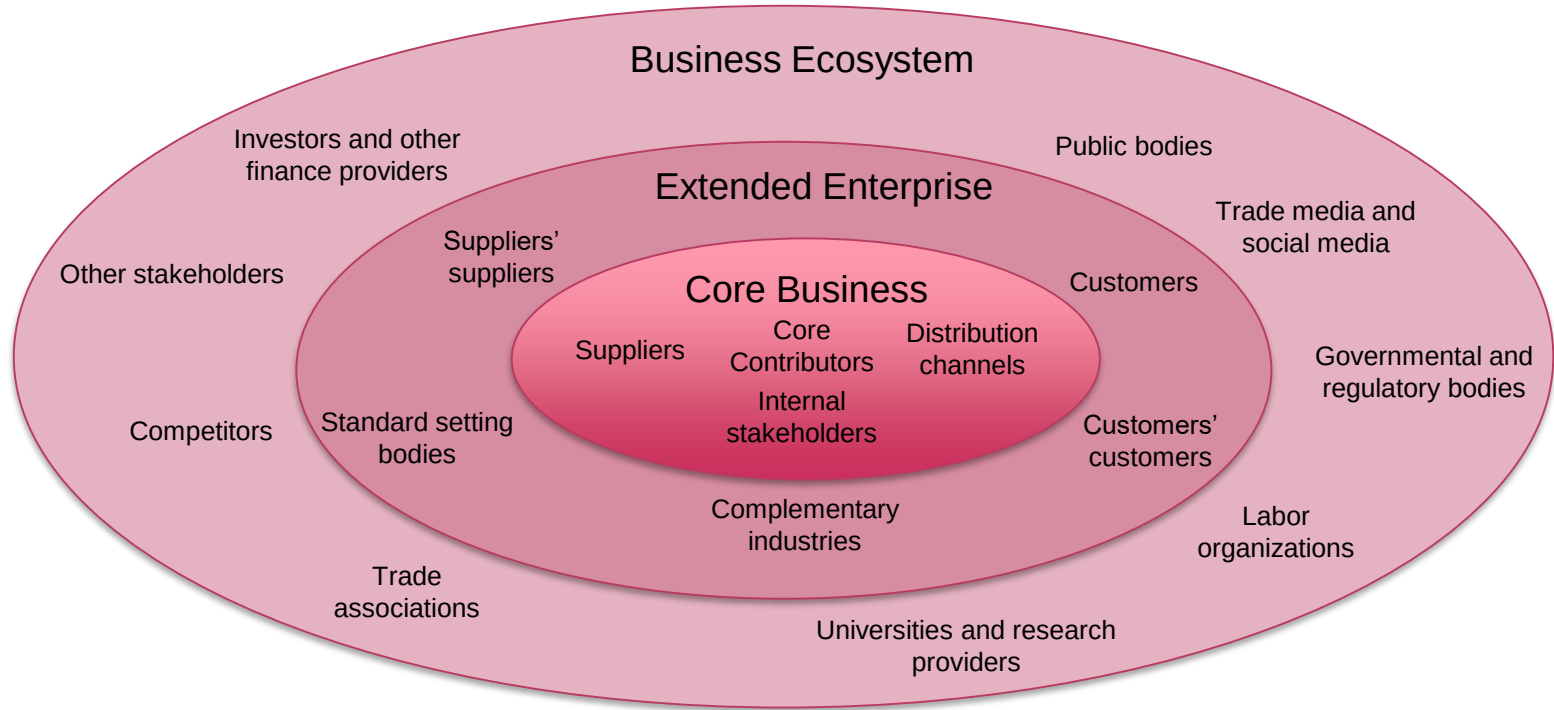
练习 1

描述锂矿公司运营所在的生态系统和商业环境中的关键要素。

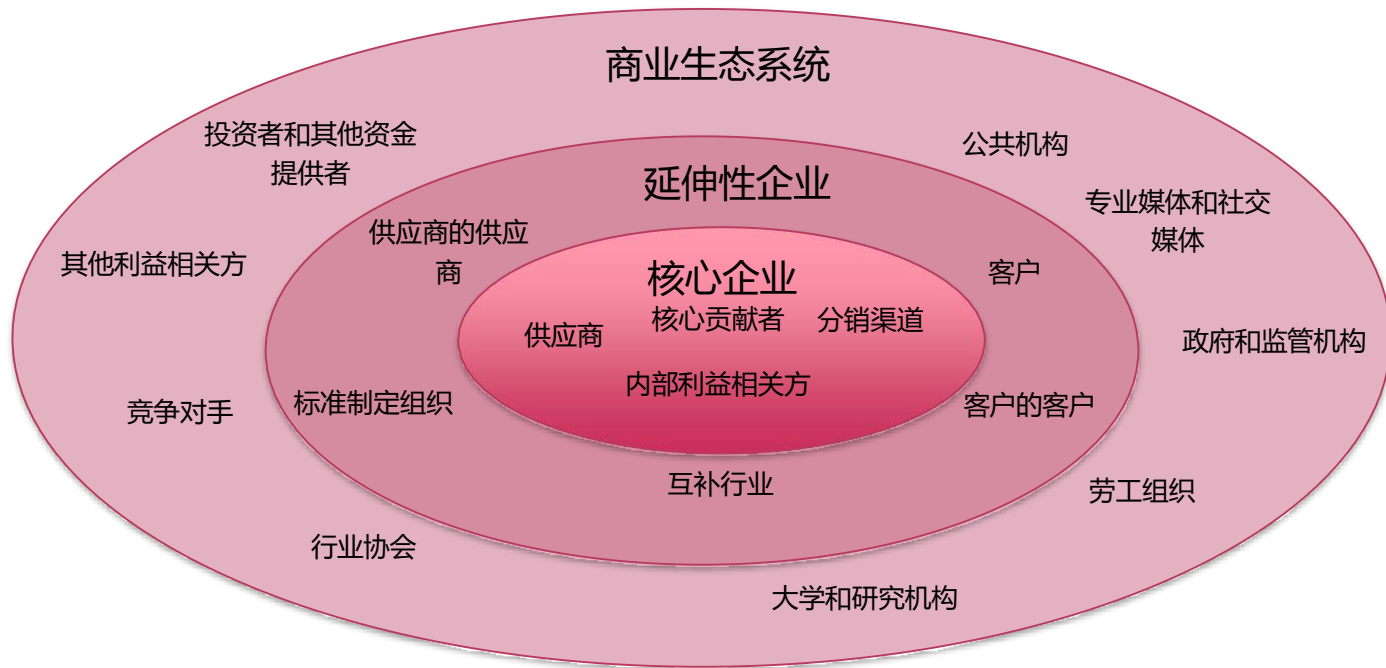
战略案例考试蓝图: B1

“我可以选择并应用适当的战略分析工具。”

Ecosystem



生态系统



Ecosystem

A network of organisations (suppliers, distributors, customers, competitors, government agencies, etc.), who are involved in the delivery of a specific product or service through both competition and cooperation.

Business ecosystems are broad by nature, potentially spanning multiple geographies and industries, including public and private institutions, and consumers. Organisations will therefore come together to **create value**.

生态系统

通过竞争与合作参与提供特定产品或服务的组织（供应商、分销商、客户、竞争者、政府机构等）网络。

商业生态系统具有广泛性，可能跨越多个地域和行业，包括公共和私营机构以及消费者。因此，各组织将共同创造价值。

Exercise 1: suggested response

Ecosystem

- Customers (e.g. battery manufacturers)
- Government licensing agencies
- Heavy duty mining vehicle/equipment manufacturers
- Hardware and software developers/vendors
- Safety equipment producers/suppliers
- Uniform designers/producers/suppliers
- Training providers, universities, researchers
- Regulatory standards organisations
- Extended supply chain and trade (e.g., fuel, parts suppliers)

练习 1: 参考答案

生态系统

- 客户（如电池制造商）
- 政府许可机构
- 重型采矿车辆/设备制造商
- 硬件和软件开发商/供应商
- 安全设备生产商/供应商
- 制服设计师/生产商/供应商
- 培训提供商、大学、研究人员
- 监管标准组织
- 扩展供应链和贸易（如燃料、零件供应商）

PESTEL Analysis

Political

- Political stability and the rule of law
- Security
- Taxation policies
- Policies on competition
- Policies on inward investment

Legal

- Laws and legal framework
- Proper IP protection
- Labour laws
- Anti-trust and competition laws

Environmental/ecological

- Weather and climate
- Environmental awareness and policies
- Natural resource availability
- Geographical location, accessibility, ease of transportation

PESTEL

Economic

- Rate of economic growth
- GDP per capita
- Interest rates
- Inflation rates
- Exchange rates
- Availability of capital

Social

- Demographics including growth of population, age structure, locations of main areas of population density
- Social culture and lifestyle
- Basic and higher education
- Human Development Index
- Social safety and benefits

Technological

- Level of technological development
- Availability of core technologies (electricity, communications, water)
- Skilled resources to service technology
- Levels of innovation and R&D
- Acceptance of new technologies

PESTEL分析

政治

- 政治稳定与法治
- 安全
- 税收政策
- 竞争政策
- 对内投资政策

法律

- 法律和法律框架
- 适当知识产权保护
- 劳动法
- 反垄断和不正当竞争法律

环境/生态

- 天气和气候
- 环保意识和政策
- 自然资源可用性
- 地理位置、可达性、交通便利度

PESTEL

经济

- 经济增长率
- 人均国民生产总值
- 利率
- 通货膨胀率
- 汇率
- 资金可用性

社会

- 人口统计数据, 包括人口增长、年龄结构、人口密度主要区域的位置
- 社会文化和生活方式
- 基础教育和高等教育
- 人类发展指数
- 社会安全和福利

技术

- 技术发展水平
- 核心技术 (电力、通信、水) 可用性
- 服务技术的技能资源
- 创新和研发水平
- 新技术接受度

Business environment: Porter's Five Forces

Each force can be given a level of strength – High, Medium or Low.

The industry will be more attractive if the forces are weak (low).

If the forces are strong (high), the profitability of the industry is low, and divestment may be a good strategy.

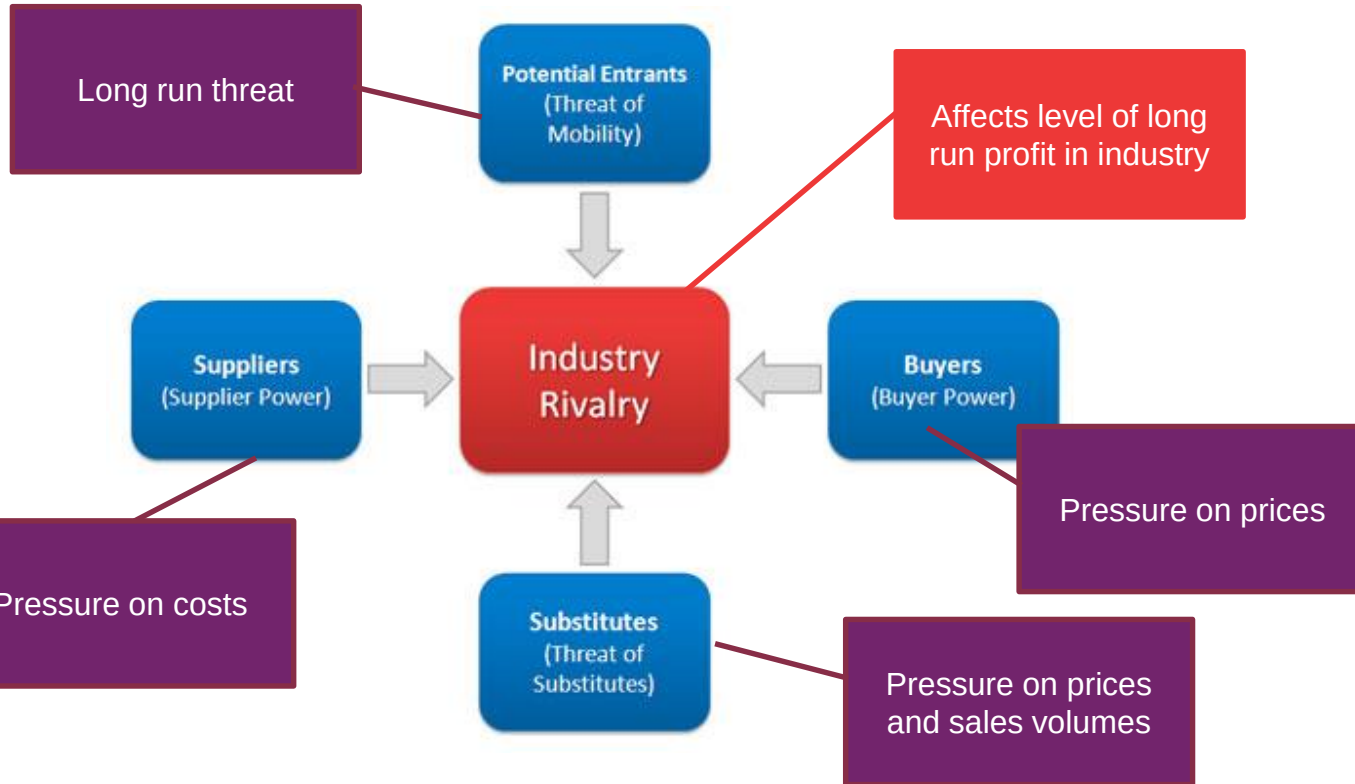
商业环境：波特的五力模型

每种力量的强度可以分为高、中、低三个等级。

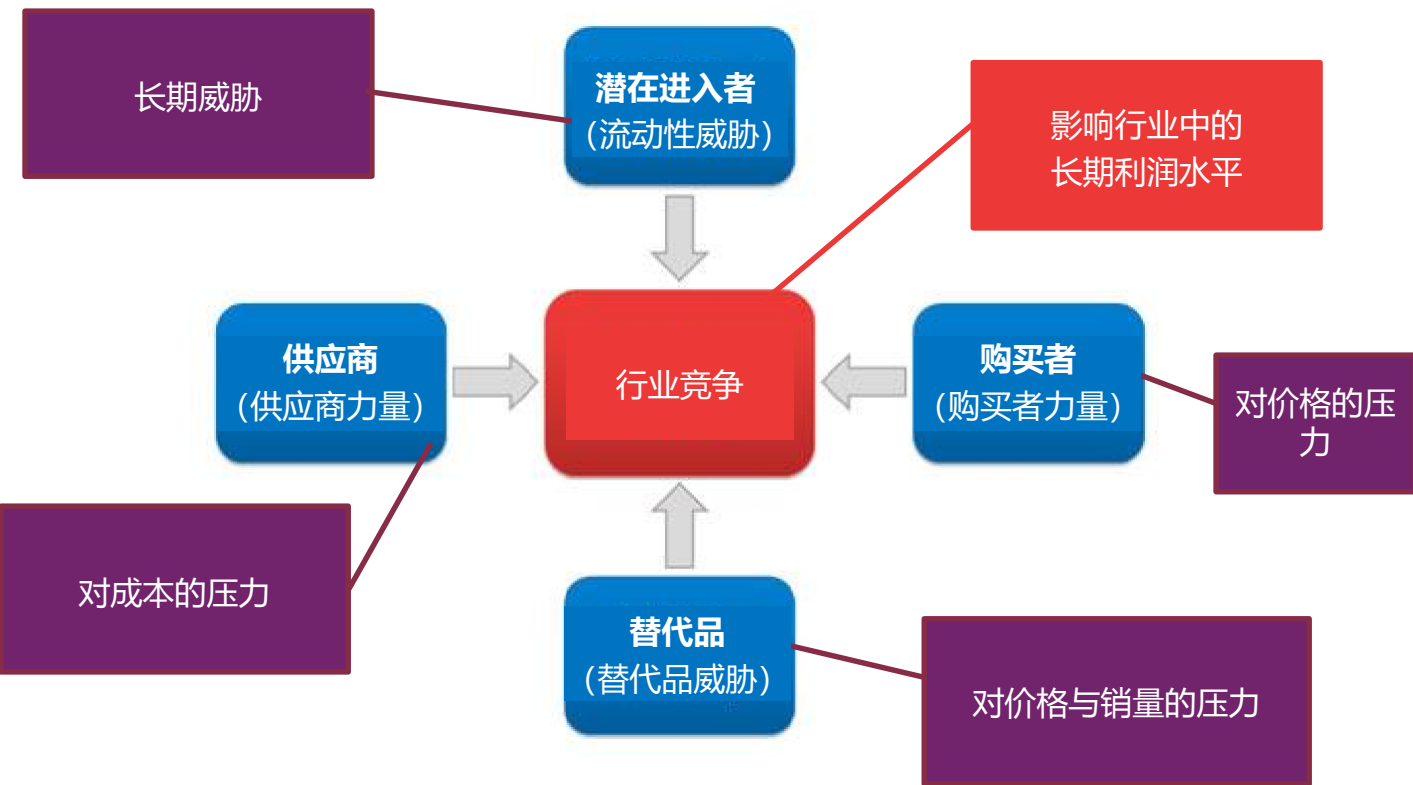
当五力较弱（低）时，行业会更具吸引力。

如果这些力量较强（高），则该行业的盈利能力较低，撤资可能是一种好的策略

Porter's Five Forces



波特五力模型



Exercise 2

Describe three main stakeholders, including their interest and potential power, in a decision by Rotomyne to diversify into battery manufacturing.

练习 2

描述罗托矿业决定多元化进入电池制造领域时的三大主要利益相关者，包括他们的利益和潜在影响力。

Stakeholders

Persons or groups with an interest in what Rotomyne does or intends to do.

- Interest: have goals or interests that may be affected by the decisions of Rotomyne.
 - Power: have the ability to help or hinder the success of what Rotomyne does or intends to do (strategy).
-

利益相关方

对罗托矿业的行为或计划感兴趣的个人或群体。

- 利益：他们的目标或利益可能会受到罗托矿业决策的影响。
 - 权力：他们有能力帮助或阻碍罗托矿业的战略或计划的成功。
-

Interest and Power

Stakeholder	Interest	Power
Employees	• Job security • Pay and conditions • Involvement	• Support management • Strike/disrupt • Leave
Managers	• Autonomy/challenge • Pay and profit share • Career progression	• Make decisions • Lead their teams • Take responsibility
Investors	• Financial return • Security of investment	• Appoint board • Buy/sell shares
Suppliers	• Revenues (price and quantity) • Consultation	• R&D support/investment • Decline to supply

利益与权力

利益相关者	利益	权力
员工	• 工作保障 • 薪酬和条件 • 参与	• 支持管理 • 罢工/破坏 • 休假
管理人员	• 自主权/挑战 • 薪酬和利润分享 • 职业发展	• 做出决策 • 领导团队 • 承担责任
投资者	• 财务回报 • 投资安全	• 任命董事会成员 • 买卖股票
供应商	• 收入（价格和数量） • 咨询	• 研发支持/投资 • 供应下降

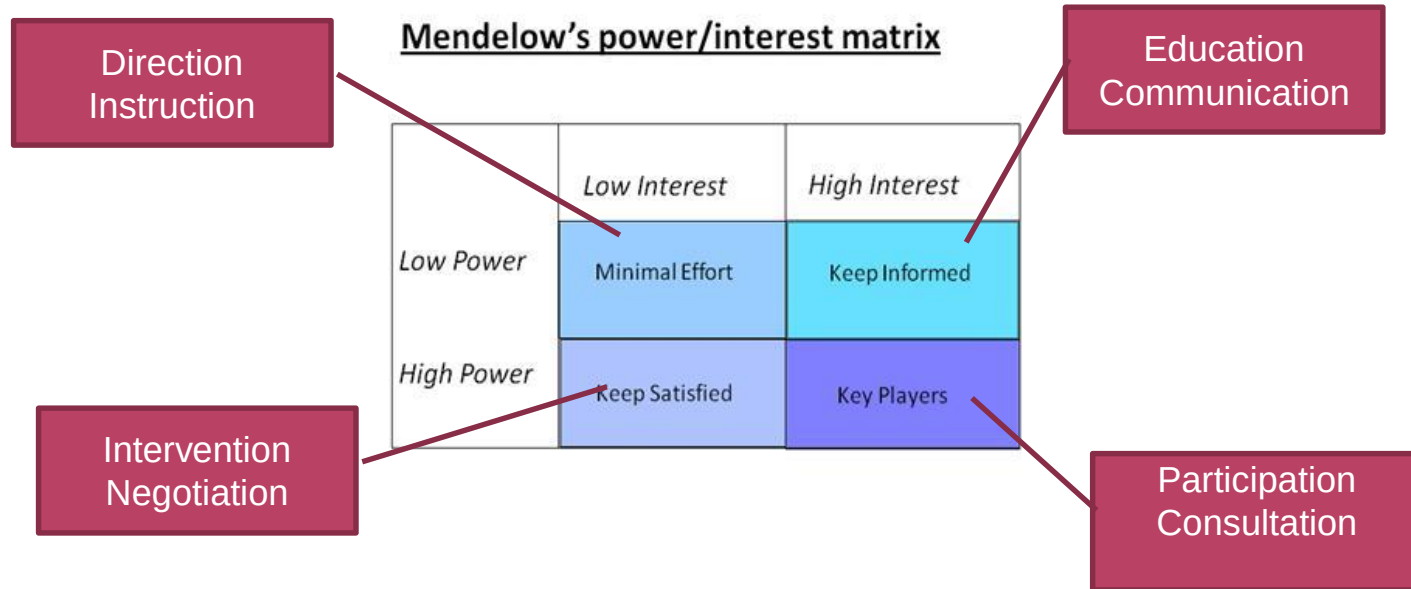
Interest and Power

Stakeholder	Interest	Power
Customers & clients	• Quality • Fair price • Service	• Purchase • Not purchase • Publicity and comment
Government bodies	• Tax revenues • Employment • Contribution to the economy and life of the country	• Legislation • Licenses • Taxation • Buying the product
Local community	• Quiet life and clean air • Employment and jobs	• Influence politicians • Direct action
Interest groups	• Promoting and protecting the things they believe are important	• Influencing public debate • Direct action

利益与权力

利益相关者	利益	权力
客户&顾客	• 质量 • 价格合理 • 服务	• 购买 • 不购买 • 宣传和评论
政府机构	• 税收收入 • 就业 • 对国家经济和生活的贡献	• 立法 • 许可证 • 税收 • 购买产品
当地社区	• 宁静的生活和清新的空气 • 就业和工作	• 影响政治家 • 直接行动
利益集团	• 促进和保护他们认为重要的事物	• 影响公众辩论 • 直接行动

Stakeholder Mapping



利益相关者分布图

Mendelow的权力/利益矩阵



Understanding Rotomyne's Business Model

理解罗托矿业商业模式

Page 7 – 9

History and strategic positioning

- Founder in 1950, owned and operated a single large open pit lithium mine, supplying lithium to wide range of industrial customers
 - Rotomyne grew steadily (organically?)
 - (strength) Survived and thrived for 70 years, started with legacy/'non-battery' customers (alloys, lubricants etc.)
 - (Strength) Experience (success/failure stories), explicit/implicit knowledge, strategy, culture
 - Became a quoted company in 1972
 - (Strength) 50 years as a quoted company: prestige, security, funding.
-

第 7 – 9页

历史与战略定位

- 成立于1950年，最初拥有并运营一个大型露天锂矿，为众多工业客户提供锂产品。
 - 罗托矿业稳步增长（是否为有机增长？）
 - （优势） 成立70年并持续繁荣，从传统或“非电池”客户开始（如合金、润滑剂等）。
 - （优势） 拥有丰富的经验（成功与失败的案例）、显性与隐性知识、战略和企业文化。1972年成为上市公司。
 - （优势） 拥有50年上市公司历史：声誉、稳定性和资金保障。
-

Page 7 – 9

History and strategic positioning

- Operates in 6 countries (6 hard rock mines, 3 traditional brine mines)
 - One of the 4 largest companies (ships products to more than 90 countries)
 - (Threat) Exposure to currency risks and wider variety of issues (political, legal, regulatory, ethical, economic, environmental etc.) in each country
 - (Threat/Risks) Supply chain: lithium processed onsite or transport ore to a different location?
-

第7 – 9

历史与战略定位

- 在6个国家运营（6个硬岩矿和3个传统卤水矿）。
 - 是全球四大锂矿公司之一，产品销往超过90个国家。
 - （威胁） 面临货币风险，以及每个国家不同的政治、法律、监管、伦理、经济、环境等多方面问题。
 - （威胁/风险） 供应链问题：锂是否在现场加工，还是将矿石运至其他地点进行加工？
-

History and strategic positioning

- Acquired the newest mine in 2004 (20 years ago) and unlikely to acquire any further new mines.
 - Geological surveys are critical.
 - (Threat) How to grow? Acquisition, vertical integration, other products (Sodium-ion / Solid-state / Lithium-Sulphur batteries)
 - (Risk) Critical to get right staff, equipment, information, research etc. – mistakes will be very costly!
-

第7 – 9页

历史与战略定位

- 于2004年收购了最新的矿山（20年前），不太可能再收购新的矿山。
 - 地质调查至关重要。
 - （威胁） 如何实现增长？并购、垂直整合、或拓展其他产品（如钠离子电池/固态电池/锂硫电池）。
 - （风险） 招募合适的员工、获取合适的设备、信息和研究成果至关重要——犯错将付出高昂代价！
-

Page 7 – 9

History and strategic positioning

- Difficult to find viable new mines, only few promising sites available for sale, potential mines are likely to offer low quality ore or brine that will be expensive to process.
- Many governments are reluctant to permit new mining operations, due environmental damage.
- (Threat) since 2004 no new mine! Growth potential is less? Less revenue by Rotomyne and competitors sue to lack of lithium?
- 7 mines in stable and well-developed countries, most recent acquisitions, in relatively high-risk countries that lack political and economic stability.
- PESTEL – political, ethical, economic ... factors

第7-9页 历史与战略定位

- 难以找到可行的新矿山，只有少数有前景的矿区可供出售，潜在矿山的矿石或盐水质量较低，加工成本高昂。
 - 许多政府因环境破坏问题而不愿批准新的采矿作业。
 - （威胁）自2004年以来没有新矿山！增长潜力减小？由于锂资源短缺，罗托矿业和竞争对手的收入减少？
 - 目前已有7座矿山位于稳定且发达的国家，但最近的收购集中在政治和经济不稳定的高风险国家。
 - 分析——政治、伦理、经济……因素。
-

Page 7 – 9

History and strategic positioning

- Head office in Porrrland, easy to manage multinational business.
- Head office provides strategic planning and oversight for the company's mining operations.
- The mines and their associated processing plants are managed locally.
- (Strength) Divisional, decentralized structure – easier to manage with local knowledge, culture and work practices.
- (Risk) Strategic drift (non conformance to mission, vision, values, strategies, objectives etc.)

第 7 – 9页

历史和战略定位

- 总部位于波兰德，便于管理跨国业务。
 - 总部负责公司采矿业务的战略规划和监督。
 - 矿山及其相关加工厂由当地管理。
 - （优势） 采用分部化、分权结构——结合当地的知识、文化和工作实践，管理更为便捷。
 - （风险） 战略偏离（未遵循使命、愿景、价值观、战略和目标等）。
-

Page 7 – 9

History and strategic positioning

- Reputation for innovation (developing processes for rapid and reliable extraction and delivery of lithium products)
 - Research and training facility (associated with a leading university), to develop new ways to use.
 - Some of the findings have led to commercial applications.
 - Collaborates with customers to development of improved batteries and other products.
 - Provide training in the safe handling and management of lithium.
-

第 7 – 9 页

历史与战略定位

- 以创新著称（开发快速可靠的锂产品提取和交付流程）。
 - 拥有与一所顶尖大学相关联的研究和培训设施，致力于开发锂的新用途。部分研究成果已转化为商业应用。
 - 与客户合作，开发改进的电池及其他产品。
 - 提供有关锂的安全处理和管理的培训。
-

Page 7 – 9

History and strategic positioning

- (Opportunity) More research with the universities to make DLE commercially viable.
 - (Opportunity) Easy to diversify into battery manufacturing.
 - (Opportunity) Pipeline for the talent
 - (Opportunity) CSR benefit (develop new products to make lives better)
-

第 7 – 9

History and strategic positioning

- （机会）加强与大学的合作研究，使直接锂提取（DLE）具备商业可行性。
 - （机会）轻松实现业务多元化，进入电池制造领域。
 - （机会）人才储备管道。
 - （机会）企业社会责任（CSR）效益（开发新产品以改善生活）。
-

Page 7 – 9

History and strategic positioning

Products	Application	% of total revenue
Battery grade lithium hydroxide and lithium carbonate	Batteries for electric vehicles and electronic devices	45%
Non-battery grade lithium hydroxide	High temperature grease for use in engines	26%
Butyllithium	Tyres and pharmaceutical products	23%
High purity lithium metal	Alloys used by aircraft manufacturers	6%
		100%

- Approx. 70% of lithium is for battery manufacturing, but 45% of revenue.
 - 49% of revenue (P\$3,829 mn) are from non-battery products (more profitable)
 - (Strength) Big, powerful customers
-

第 7 – 9 页

历史与战略定位

罗托矿业生产的产品如下：

产品	应用	占总收入的百分比
电池级氢氧化锂和碳酸锂	电动汽车和电子设备的电池	45%
非电池级氢氧化锂	用于发动机的高温润滑脂	26%
丁基锂	轮胎及医药产品	23%
高纯锂金属	飞机制造商使用的合金	6%
		100%

- 约70%的锂用于电池制造，但仅占收入的45%。
- 49%的收入（P\$38.29亿）来自非电池产品（利润更高）。
- （优势） 拥有大型且有实力的客户。

Page 7 – 9

History and strategic positioning

- The mines are all highly automated, reducing the need to expose workers to dust and fumes.
 - Operations are managed centrally; tracks inventory levels and anticipated shipments of products.
 - (Risk) Keep supply steady to sustain business and mitigate cashflow and working capital risks
 - Largest customer (major vehicle manufacturer) 22% of revenue (P\$1,719)
 - (Risk) High buyer power, key player – may move to a competitor?
 - (Threat) not necessarily an EV manufacturer
-

第 7 – 9

历史与战略定位

- 矿山高度自动化，减少了工人暴露在粉尘和烟雾中的风险。
 - 运营由总部集中管理，监控库存水平和预期产品出货量。
 - （风险） 保持供应稳定以维持业务运营，并缓解现金流和营运资本风险。
 - 最大的客户（主要汽车制造商）占收入的22%（P\$17.19亿）。
 - （风险） 买方议价能力强，作为关键客户，可能转向竞争对手？
 - （威胁） 不一定是电动汽车制造商。
-

Exercise 3

How could Rotomyne grow its business? (use the Ansoff's Matrix to help you to generate ideas)

练习3

罗托矿业如何实现业务增长？（可利用安索夫矩阵来帮助产生新思路）

Ansoff Matrix

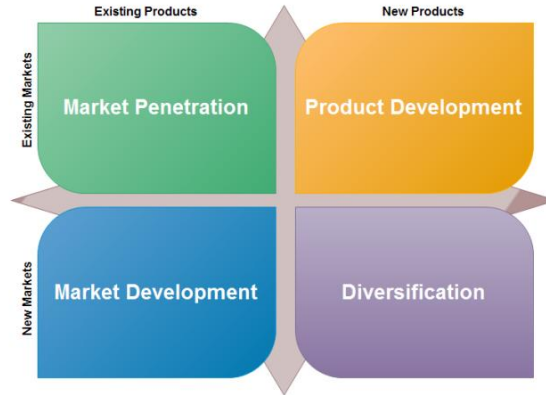
Increase sales and market share

- Greater promotion
- Expand capacity
- More distribution channels

Launch new products

- Complements for existing products
- Something the customer buys from another business
- A breakthrough technology

Product-Market Ansoff Matrix



Expand the firm into new markets

- New customer segments
- New uses for product
- New countries

Enter new industries

- Vertical integration
 - Technology extension
 - Risk avoidance
-

安索夫矩阵

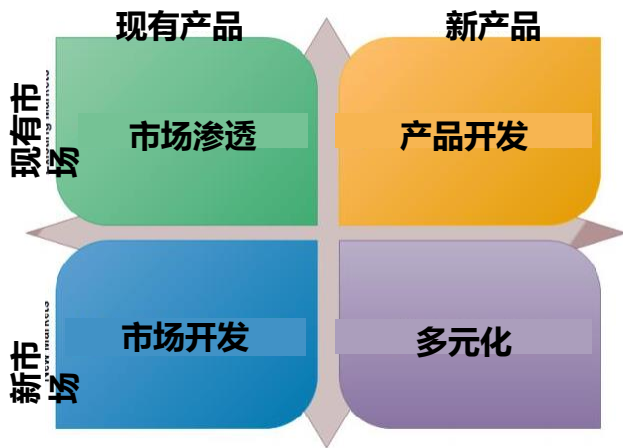
提高销售额与市场份额

- 加强促销
- 扩大产能
- 增加分销渠道

发布新产品

- 现有产品的互补产品
- 客户从其他企业购买的产品
- 突破性技术

安索夫产品-市场矩阵



企业扩张到新市场中

- 新客户细分市场
- 产品新用途
- 新国家/地区

进入新行业

- 垂直整合
- 技术延伸
- 规避风险

Exercise 4

How may Rotomyne use Big Data and Analytics?



练习4

罗托矿业如何利用大数据和分析技术？



Volume 体量
Size of Data



Validity 有效
Data quality, Governance, Master Data Management on Massive



Velocity 高速
The Speed at which Data is Generated



Variability 可变性
Dynamic, Evolving Behavior in Data Source



Variety 多样
Different type of Data



Venue 场地
Distributed Heterogeneous Data from Multiple Platforms



Veracity 真实
Data Accuracy



Vocabulary 词汇
Data Models, Semantics that describes data Structure



Value 价值
Useful Data



Vagueness 模糊
Confusion over Meaning of BigData and Tools used



Characteristics of Big Data

Characteristics of Big Data

Volume:

Vast amounts too large to be stored and analysed by conventional data bases. Need for distributed processing (e.g. cloud)

Variety:

Unstructured e.g. messages, social media postings, photos, sensor data, video and voice recordings.

Velocity:

Constantly generated and rapidly changing. Requires real-time capture and analytics.

Veracity:

Accuracy of data - may be more true due to large numbers.

Value:

Can provide huge business benefits.

大数据的特点

大数据特点

体量：

海量数据，难以通过传统的数据库存储与分析。需要分布式处理（如云技术）

多样：

非结构性，如短信、社交媒体帖子、照片、传感器数据、视频和录音

高速：

不断生成，快速变化。需要实时捕捉与分析

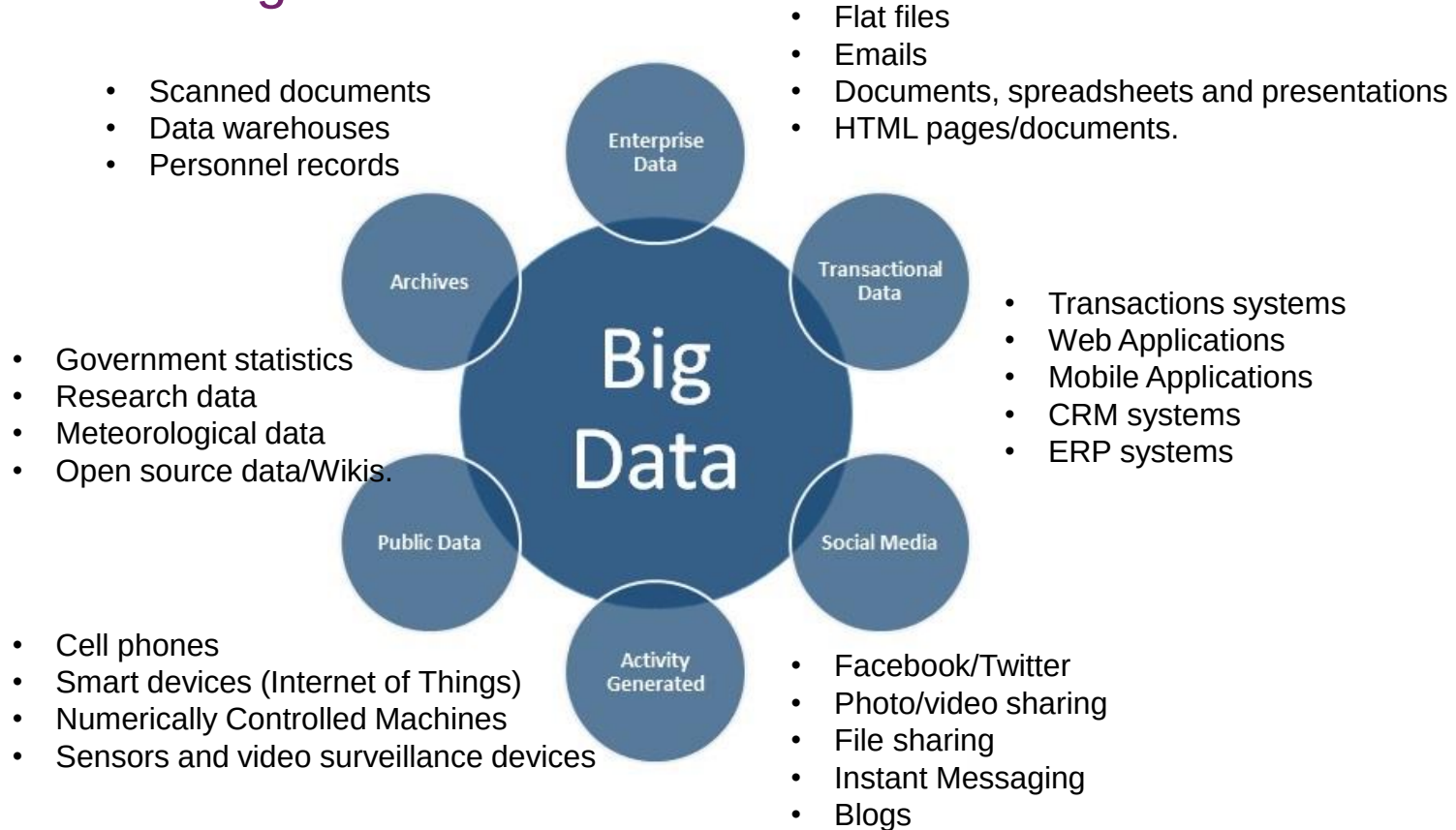
真实性：

由于体量庞大，可能更为真实

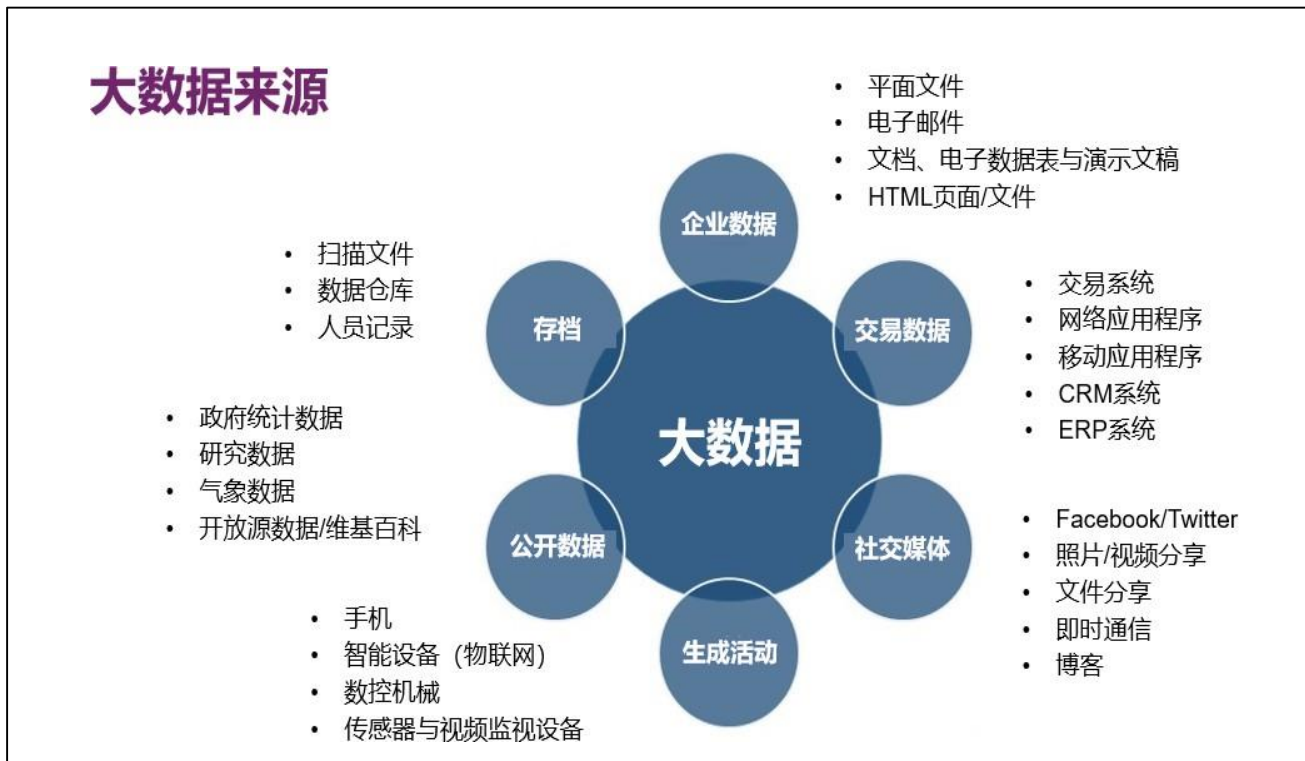
价值：

可提供巨大的商业效益

Sources of Big Data

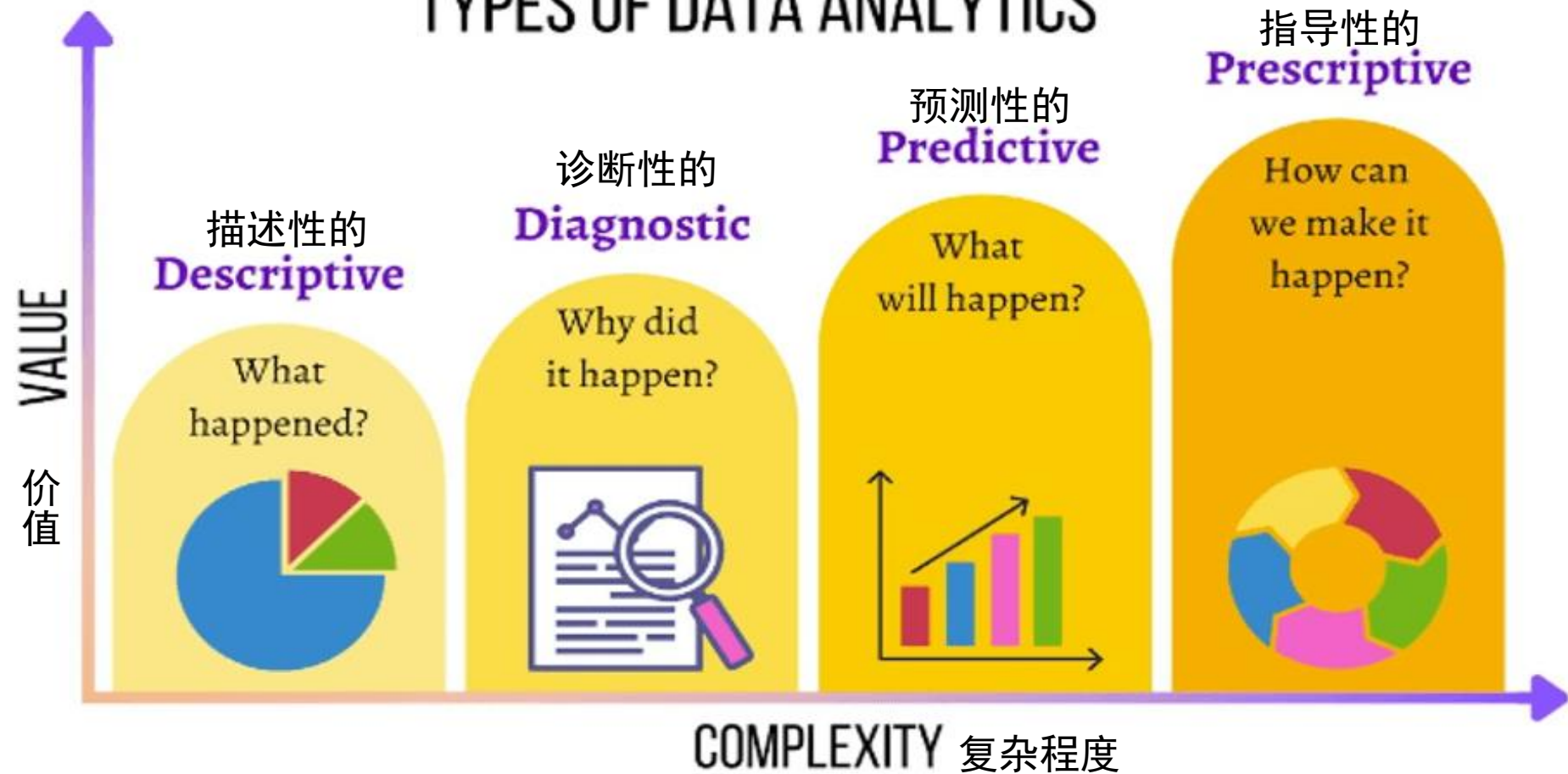


大数据来源



数据分析类型

TYPES OF DATA ANALYTICS



Big Data Analytics

Descriptive Analytics	Examples
• Associations between data items	• Items commonly bought together • Weather/season and store footfall • Personal characteristics and illness/mortality • Components/suppliers and product failures
• Clusters of data items	• Customer segments/types • Hot spots of illnesses • Slow moving inventory items/locations • Peer networks in the organisation
Predictive Analytics	Examples
• Sequences in data items	• Symptoms that are lead indicators for an illness • Customer actions that may be followed by loss of the customer • Route followed giving guide to final destination
• Trends in data items	• Increasing/ decreasing popular interest in something • Changes in prices of materials or foreign currency • Warranty claims by cause rising indicating a manufacturing defect

大数据分析

描述性/跨行业	示例
数据项之间的联想	- 经常一起购买的商品/服务 - 气候/季节与门店客流量 - 个人特点与疾病/死亡率 - 部件/供应商与产品故障
数据项聚类	- 消费者细分市场/类型 - 疾病热点 - 滞销库存商品/地点 - 组织中的同级网络
预测性/时序	示例
数据项的顺序	- 作为疾病领先指标的症状 - 客户流失后可能发生的客户行动 - 所遵循的路线提供最终目的地的导航
数据项的趋势	- 对某个事物的兴趣上升/下降 - 原料价格或汇率变动 - 因代表制造缺陷的原因而产生的保修申请

Big Data Analytics

Prescriptive Analytics	Examples
Identifies the best course of action or decision given the known parameters	- Suggesting interventions to deal with illness - Likely future prices of commodities - Automated share dealing - Fault finding in a piece of equipment - Best tactics in a competitive game - Where to drill for oil - Optimizing layout and product range in a store - Optimizing pricing of a product or service (dynamic pricing) - Spotting bogus transactions or fake products in a supply chain

大数据分析

预测性/时序	示例
根据已知数据识别最佳的行动方案 and 决定	- 表明需要通过干预来解决患病的情况 - 可能涉及商品的未来价格 - 自动股票交易 - 查找设备的故障问题 - 竞争中的最佳策略 - 钻探石油的位置 - 优化门店布局和产品范围 - 优化产品或服务的定价（动态定价） - 识别供应链中的虚高交易或假冒产品

Rotomyne's purpose, strategy and values

- Mission is 'to power consumers' lives in a world that is clean, healthy and sustainable."
- Vision is 'to meet customers' needs for the reliable supply of good quality materials. In particular, Rotomyne wishes to transform the delivery of power through its support for customers who depend on lithium for their products."
- Values – ethically and honestly, sustainability, safe working environment, constantly innovate and responsive to customer needs.

Cambells mission statement:

Purpose – why?
Strategy – how?
Values – what?

Drive strategy and success factors ... (what are they?)

第 9 页

罗托矿业的宗旨、战略和价值观

- 使命是“在一个清洁、健康和可持续的世界中为消费者的生活提供动力。”
- 愿景是“满足客户对优质材料可靠供应的需求。尤其是，罗托矿业希望通过支持依赖锂产品的客户来改变能源的交付方式。”
- 价值观——以道德和诚实的方式行事，追求可持续发展，提供安全的工作环境，持续创新并对客户需求作出响应。

坎贝尔公司的使命声明：宗旨 – 为什么？战略 – 如何实现？价值观 – 体现了什么？

推动战略和成功因素……（这些是什么？）

第 9 页

Rotomyne's purpose, strategy and value

- SAF criteria/framework (**suitability**, acceptability and feasibility)
- Meeting customer needs (being responsive to them) – customer is a key player
- Ensuring reliable supply – supply chain management is critical
- Sustainability – more investment into R&D, DLE, other alternatives, new sites, improve extraction process and reduce wastage
- Safe working environment – invest in more automation

Cambells mission statement:

Purpose – why?
Strategy – how?
Values – what?

Drive strategy
and success
factors ... (what
are they?)

第 9 页

罗托矿业的宗旨、战略和价值观

- SAF标准/框架（适宜性、可接受性和可行性）
- 满足客户需求（对客户需求的快速响应）——客户是关键角色
- 确保可靠供应——供应链管理至关重要
- 可持续性——增加对研发、直接锂提取（DLE）及其他替代方案的投资，开发新矿区，改进提取工艺并减少浪费
- 安全的工作环境——投资更多的自动化

坎贝尔公司的使命声明：宗旨 – 为什么？战略 – 如何实现？价值观 – 体现了什么？

推动战略和成功因素……（它们是什么？）

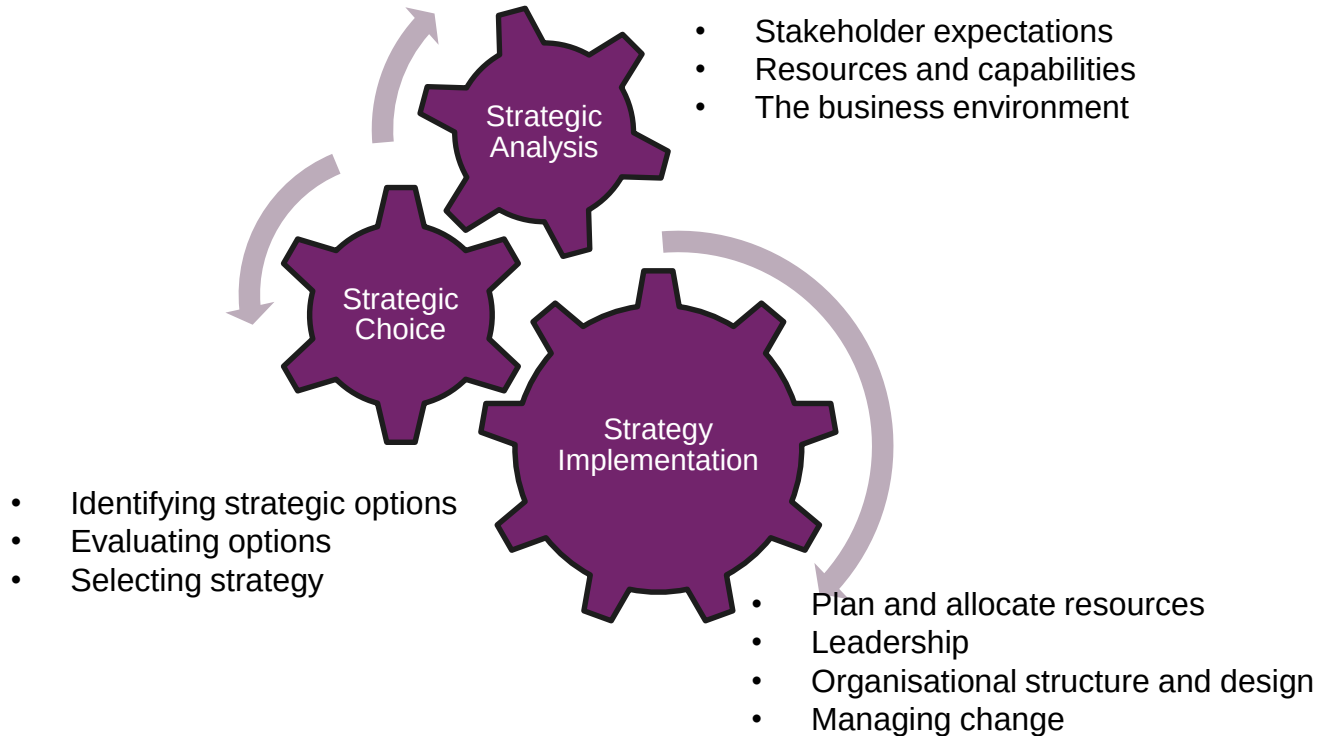
Exercise 5

How should Rotomyne's Mission Vision and Values (MV&V) affect the strategies it adopts?

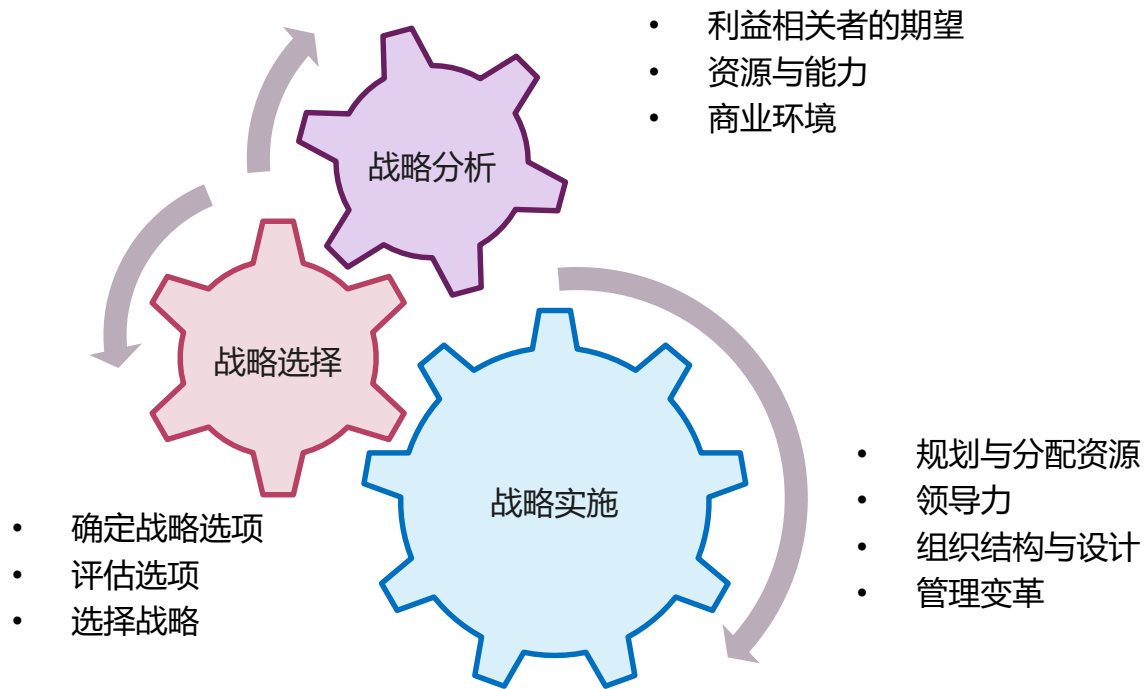
练习5

罗托矿业和使命、愿景和价值观（MV&V）应如何影响其所采用的战略？

Building a Strategy



构建战略



Exercise 5 - suggested responses

- Provides an indication of stakeholder expectations
 - MV&V may have been formulated to reflect views of senior managers and others
 - Staff, investors, and others engage with Rotomyne on the basis of its MV&V. Departing from the MV&V will cause disengagement or resistance
 - Indicates the areas of competence or strengths of the business
 - The client-relations, skills and assets of Rotomyne have been assembled to support its present MV&V
 - People at Rotomyne must be inspired to live the MV&V to create and deliver value to stakeholders.
-

练习 5 – 参考答案

- 提供了对利益相关者期望的指示。

- 罗托矿业和使命、愿景和价值观（MV&V）可能是为了反映高管层和其他关键人物的观点而制定的。

- 员工、投资者等根据其 MV&V 与罗托矿业互动。背离 MV&V 可能导致参与度降低或出现抵抗。

- MV&V还展示了企业的核心竞争力或优势所在。

- 罗托矿业的客户关系、技能和资源都是为了支持其当前的MV&V而建立的。。

- 罗托矿业的每位员工必须被激励去践行MV&V，以为利益相关者创造并传递价值。

Exercise 5 - suggested responses

- Basis for evaluating the **suitability** of a strategy
 - Strategies that are consistent with MV&V are more favorable
 - Strategies that are not consistent with MV&V may be disruptive or may require complex change leadership to make them successful

BUT

- MV&V only important if it is accurate expression of Saefwell's values and goals and not just a public relations statement
 - MV&V must be implemented in the strategy, standards of performance and behavior, and values of Saefwell to be effective
 - MV&V must be allowed to change through time as the business environment and interests of stakeholders change
-

练习 5 – 参考答案

- 策略适宜性评估的依据：

- 与MV&V（使命、愿景和价值观）一致的策略通常更受青睐。

- 不符合MV&V的策略可能会引起混乱，或可能需要复杂的变革领导力才能成功。

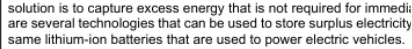
但是，

- MV&V仅在准确反映塞夫韦尔的价值和目标，而不仅仅是一份公关声明时才重要。
 - MV&V必须在策略、绩效标准、行为准则和塞夫韦尔的价值观中得到实施，才能发挥效果。
 - 随着商业环境和利益相关者利益的变化，MV&V也应该适时进行调整。
-

Review of Industry Articles

行业文章回顾

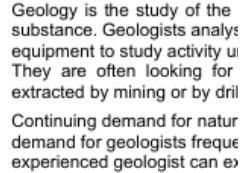
Scientists debate battery storage options



This year's annual action plan calls for carbon-free energy generation. It also draws attention on the security of energy supply. This has been a significant shift in focus for many of the alternative energy sources that can deliver power.

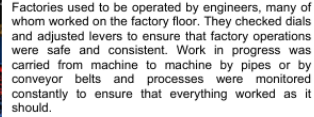
Wind and solar energy
winds blow and the skies
locations can provide

Careers - geologists are in demand



A career in geology can be challenging. There are many techniques used to explore beneath the earth's surface, but it is rarely possible to be 100% certain that a geologist may be confident of the presence of, say, iron ore, but that it is to sink a test drill to collect samples. Iron ore may be present, but the grade or the deposit may be insufficient to justify mining. There is also the earth's structure in that area is not sufficiently stable to mine safely.

OT is the new threat vector



Factories are now more likely to be controlled by Operational Technology (OT) rather than by human combination of hardware and software that monitors and equipment, processes and events in the workplace.

OT enables factories to be managed centrally, with processes being managed by automation rather than human operators. Machines are not simply automatic; they can pass instructions to adjacent equipment, activating valves and carriers to transfer materials to the next stage in the process and sending the correct settings to the machinery that will conduct the next stage of manufacture.

The Industrial Control Systems (ICS) that enable this communication and coordination are not new, they were developed decades ago. They were designed to enable safe and efficient operations, with little concern for security because there was rarely any external connectivity. What is new is that ICS is being integrated with IT systems, despite the lack of security features in ICS software. Connecting ICS systems to IT networks creates opportunities for unauthorised breaches that can be motivated by a desire to access confidential data or to disrupt manufacturing operations.

May introduce new themes that could not be fit-into the pre-seen story.

波兰德电讯报

科学家们就电池存储选择问题争论不休



今年关于无碳能源发电的年度学术会议将着重关注电力供应的安全性。这是一个值得关注的问题，因为许多化石燃料替代品所依赖的能源在输送的电力方面可能不稳定。

当有风且天空晴朗时，风能和太阳能是最优选择，但并不能向所有地方稳定供电。一种解决方案是将不需要立即使用的多余电能收集起来。有多种技术可用于存储多余电能，包括电动汽车所采用的锂离子电池阵列。

用于储能的锂离子电池的需要有限，因为将其用于储能成本过于高昂。替代电池类型正在开发中，其中一些将适用于有效存储多余电能。钠离子电池的生产成本更低，因为使用的原材料更多样。到 2030 年，应该可以实现商业化。

钠离子电池比同等的锂离子电池更大、更重，因此不适用于电动汽车。如果在有风的日子将钠离子电池放在风电场中，用于储存不需立即使用的电能，那么电池体积大在这种情况下便不成问题。

地质学人才

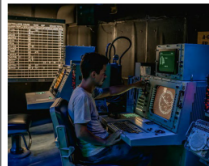
地质学是研究地球结构及其物质的学科。地要从事岩石样本的分析，并利用设备研究地动。他们经常寻找可以通过采矿或钻井提取

我们对自然资源的持续需求意味着需要大量才。经验丰富的地质学家有望获得优厚的待

从事地质学的工作颇具挑战性。尽管有很多法 100% 确定到底是什么。地质学家可能确收集样本。虽然可能存在铁矿石，但矿石的合理性。还有一种可能是，该地区的地质

波兰德电讯报

OT 是新的威胁载体



工厂过去由工程师操作，许多工程师在工厂车间工作。他们会检查刻度盘并调整杠杆，以确保工厂作业安全一致。在制品通过管道或传送带从一台机器传送到另一台机器，并不断监控过程，以确保一切正常工作。

而今，工厂更多采用操作技术(OT)来操控，而非人工干预。OT 包括硬件和软件组合，用于监视和指导工作场所中设备、流程和事件的操作和控制。

OT 实现了工厂的集中管理，流程完全自动化运行，而非由人工操作员管理。机器不仅是自动化运行；甚至可以将指令传递给相邻设备，激活阀门和载体，以将材料转移到下一阶段的工艺流程，并将正确设置发送到进行下一阶段制造的机器上。

实现这种沟通和协调的工业控制系统(ICS)并不鲜见，数十年前便已开发完成。系统的设计旨在实现安全高效的运营，但很少考虑安全问题，因为很少进行任何外部连接。与以往不同的是，尽管 ICS 软件缺乏安全功能，但其正逐步与 IT 系统集成。将 ICS 系统连接到 IT 网络会为未经授权的行为创造机会，这些违规行为可能出于访问机密数据或破坏制造运营的动机。

对潜在考试问题的洞察。思考这些问题将带来哪些机会和威胁；罗托矿业可以做些什么来应对它们？

你会在考试任务的“正在发展的故事”或（未见的）参考材料中找到这些问题的重复出现。

支持预览材料中的常见主题。

可能引入无法融入预览材料故事的新主题。

Happy Comic

Readers' questions



Question: How do they last for?

Answer: All batteries consist of three basic parts: two metal electrodes (the "cathode" and the "anode") and a liquid or paste called the electrolyte.

Connecting the battery (perhaps by plugging a battery charger, which is just another term for electric

The external flow of electrons is driven to the cathode. The ions are created through the electrolyte. The chemical

to the cathode. The ions are created by a chemical reaction inside the battery and they flow through the electrolyte. The chemical reaction inside the battery continues until the electrodes can no longer create or accept the positive ions. If the battery is single use, then it is now flat and can no longer be used as a power source.

Rechargeable batteries differ from single use because it is possible to reverse the chemical process by applying an electrical charge to the electrodes from the outside. The charge sends positive ions from the cathode back to the anode. Unfortunately, this process is not perfect. The electrodes cannot accept as many ions as were lost during their previous use as a power source, which means that the battery's capacity reduces slightly every time it is recharged.



Answer: In theory, that would be a great idea, but it isn't really practical. The electrolyte in a lithium-ion battery is held in modules that are made up of large numbers of cells. Recovering the lithium requires the cells to be opened up and emptied, which requires a lot of time and a lot of energy. It wouldn't necessarily benefit the environment to extract the lithium from old batteries instead of mining more.

(Threat) Lithium spent is not replaceable. There is limited amount of lithium on the ground.

(Opportunity) Invest in DLE alternative, recycling to make it commercially viable (Rotomyne values innovation and technology and can work with the universities)

The University of Manchester, in collaboration with Lion Vision, developed a system that uses advanced vision systems and machine-learning technologies to detect and extract lithium-ion batteries and other hazardous items from waste streams. This innovative solution system has already been implemented at waste processing sites in Kent. (UK), providing a scalable solution for managing battery waste, reducing environmental risks, and improving safety in the waste industry.

快乐动漫每一天

读者问题



问题：可充电

答：所有电池
（“负极”和“正
物，用于将两
（比如将电池
这是电的另一种

外部电子流由带正电的离子从正极向负极驱
会流经电解质。电池内部的化学反应一直持
止。如果为一次性电池，那么此时电量耗尽

可充电电池不同于一次性电池，因其可通
程。电荷将正离子从负极送回正极。遗憾的

外部电子流由带正电的离子从正极向负极驱动。离子由电池内部的化学反应产生，
会流经电解质。电池内部的化学反应一直持续到电极不能再产生或接受正离子为
止。如果为一次性电池，那么此时电量耗尽，不能再用作电源。

可充电电池不同于一次性电池，因其可通过从外部向电极施加电荷来逆转化学过
程。电荷将正离子从负极送回正极。遗憾的是，这个过程并不完美。电极无法接受
之前作为电源使用时失去的相同离子数量，这意味着每次充电时电池的容量都会略



问题：如果锂矿开采对环境有害，为什么我们不在电池无法再充电
后从电池中回收锂呢？

Mathew, 12 岁

答：从理论上讲，这是个好主意，但并不实际。锂离子电池中的电
解质保存在由大量电芯组成的模块中。回收锂需要将电芯打开并清
空，这需要耗费大量时间和能源。从旧电池中提取锂而不是开采更
多锂，不一定对环境有益。

（威胁） 锂资源耗尽后不可再生。地球上的锂储量有限。（机会） 投资于直接锂提取（DLE）替代方案和回
收技术，使其具备商业可行性（罗托矿业重视创新与技术，且可以与大学合作）。

曼彻斯特大学与Lion Vision合作开发了一种系统，利用先进的视觉系统和机器学习技术从废物流中检测并提取锂离子电池和其他
危险物品。这一创新解决方案已在英国肯特郡的废物处理场实施，为电池废物管理提供了可扩展的解决方案，减少了环境风险并
提高了废物处理行业的安全性。

Happy Comic

Readers' questions



Question:
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Answer: /
electrodes
called the
Connecti

battery (perhaps by plugging a batte
which is just another term for electrici

The external flow of electrons is drive
to the cathode. The ions are created i
through the electrolyte. The chemi

It does not matter where lithium was mined once it has been processed. Lithium carbonate produced in one country is identical to lithium from any other. It is sold as a commodity and the price of lithium as a raw material is the same regardless of its source.

Location can make a difference to the mining companies because their operating costs will be affected by the purity of the mined rocks or brine. If the material from the mine is of a higher purity, then it will be cheaper and easier to extract the lithium carbonate. Location also affects transportation costs.



Question: We are still learning about batteries in chemistry. We found out today that the electrolyte used in lots of rechargeable batteries is either lithium carbonate or lithium hydroxide. They still get called "lithium" even though they are compounds of the pure lithium metal. Isn't that confusing?

Jiayi, age 11

Answer: It is common practice to refer to different compounds of lithium in this way. For most purposes, the context in which the word is used makes its meaning clear. It is possible to extract lithium in its pure metal form, but it is difficult to store because it is highly volatile and deteriorates quickly if it comes into contact with the atmosphere. It is unlikely that any mention of "lithium" is referring to pure lithium metal.

(Threat) Cannot differentiate as lithium is sold as a commodity. Economy and market impact the price.

(Opportunity) Steady supply is better for efficiency, reliability and profit. Invest to find mines with more purity or extract more % of lithium from ore or brine.

(Opportunity) Could we differentiate and meet (psychological) needs better?

快乐动漫每一天

读者问题



问题：可

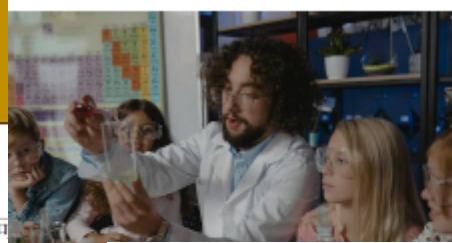
答：所有

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外部电子流由带正电的离子从正极向负极驱会流经电解质。电池内部的化学反应一直持续。如果为一次性电池，那么此时电量耗尽

可充电电池不同于一次性电池，因其可通过程。电荷将正离子从负极送回正极。遗憾的

理经过
不同。

这是电

其所指含义。虽然可以提取纯金属形式的锂，但因其极易挥发，导致很难储存，一旦与大气接触就会迅速变质。提到“锂”时，不太可能是指纯金属锂。

显然，采购锂化合物（或纯锂金属）的公司会非常具体地说明所订购的材料，因此不会出现混淆。

问题：我们一直从化学方面研究电池。我的一个同学说她的祖国出产世界上最好的锂。这是真的吗？

Jiayi, 11 岁

答：否。锂开采自多个不同的国家。它来自地球，以含有锂化合物的岩石或卤水的形式存在。岩石和卤水经过

问题：我们目前仍从化学方面研究电池。我们发现，目前许多可充电电池中使用的电解质要么是碳酸锂，要么是氢氧化锂。尽管属于纯锂金属化合物，但它们仍然被称为“锂”。这不令人困惑吗？

无

Jiayi, 11 岁

答：用这种方式来指代不同的锂化合物是很常见的做法。在大多数情况下，通过“锂”一词所处的上下文，即可明确

（威胁） 锂作为一种商品出售，无法实现差异化。经济和市场因素影响价格。

（机会） 稳定的供应有助于提高效率、可靠性和利润。可以投资寻找更高纯度的矿山，或提高从矿石或卤水中提取锂的比例。

（机会） 我们是否可以通过差异化来更好地满足客户的（心理）需求？

Portland Telegraph

OT is the new threat vector



Factories used to be places where workers were safe and adjusted to the environment. They were safe and carried on conveyor belts constantly to or from the factory.

Factories are now places where workers are not safe and adjusted to the environment. They are not safe and carried on conveyor belts constantly to or from the factory.

OT consists of the combination of hardware and software that directs the operation and control of equipment, processes, and machines.

OT enables factories to be managed centrally, with processes being managed by automation rather than human operators. Machines are not simply automatic; they can pass instructions to adjacent equipment, activating valves and carriers to transfer materials to the next stage in the process and sending the correct settings to the machinery that will conduct the next stage of manufacture.

The Industrial Control Systems (ICS) that enable this communication and coordination are not new, they were developed decades ago. They were designed to enable safe and efficient operations, with little concern for security because there was rarely any external connectivity. What is new is that ICS is being integrated with IT systems, despite the lack of security features in ICS software. Connecting ICS systems to IT networks creates opportunities for unauthorised breaches that can be motivated by a desire to access confidential data or to disrupt manufacturing operations.

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Expensive to set-up and technology is changing fast.

(Opportunity) Show our commitment to innovate and ensure high quality, efficiency and reduced wastage.

(Threat/Risk) Integrate cyber-security with overall risk management. Up to date hardware, software, networking, and trained staff (with right mindset) are critical.



工厂过去由工程师操作，许多工程师在工厂车间工作。他们会检查刻度盘并调整杠杆，以确保工厂作业安全。在制品通过管道或传送带从一台机器传送到另一台机器，并不断监控过程，以确保一切正常工作。

而今，工厂更多采用操作技术(OT)来操控，而非人工干预。OT包括硬件和软件组合，用于监视和指导工厂场所中设备、流程和事件的操作和控制。

OT实现了工厂的集中管理，流程完全自动化运行，而非由人工操作员管理。机器不仅是自动化运行；甚至可以将指令传递给相邻设备，激活阀门和载体，以将材料转移到下一阶段的工艺流程，并将正确设置发送到进行下一阶段制造的机器上。

实现这种沟通和协调的工业控制系统(ICS)并不鲜见，数十年前便已开发完成。系统的设计旨在实现安全高效的运行，但很少考虑安全问题，因为很少进行任何外部连接。与以往不同的是，尽管 ICS 软件缺乏安全功能，但其正逐步与 IT 系统集成。将 ICS 系统连接到 IT 网络会为未经授权的违规行为创造机会，这些违规行为可能出于访问机密数据或破坏制造运营的动机。

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成本高) 建立系统成本昂贵，且技术发展迅速。

(机会) 展示我们对创新的承诺，确保高质量、高效率并减少浪费。

(威胁/风险) 将网络安全整合到整体风险管理中。保持硬件、软件和网络系统的更新，以及培训具备正确心态的员工至关重要。

Portland Telegraph

Scientists debate battery storage



This year carbon-attention. This has many of energy delivery

Wind a winds b location

solution is to capture excess energy that is not required for immediate consumption. There are several technologies that can be used to store surplus electricity, including arrays of the same lithium-ion batteries that are used to power electric vehicles.

Demand for lithium-ion batteries for energy storage is limited because the batteries are too expensive for this purpose. Alternative battery types are under development, some of which will be suitable for the efficient storage of surplus electricity. Sodium-ion batteries are much cheaper to produce because they use more abundant materials. They should be commercially available by 2030. They are larger and heavier than equivalent lithium-ion batteries, making them unsuitable for use in electric vehicles. That greater bulk is not a problem if the batteries are to be located in a windfarm as a means of storing electricity that is not required for immediate consumption, perhaps on a windy day.

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Common themes: Limited lithium deposits, difficult to get government permission, environmental-unfriendly and costly process

(Opportunity) Diversify into Sodium mining, Sodium-ion battery manufacturing, etc.

波兰德电讯报

科学家们就电池存储选择问题争论不休



今年关于无碳能源发电的年度学
关注电力供应的安全性。这是一
问题，因为许多化石燃料替代品
在输送的电力方面可能不稳定。

当有风且天空晴朗时，风能和太
择，但并不能向所有地方稳定供
方案是将不需要立即使用的多余
有多种技术可用来储存多余电能。
车所采用的锂离子电池阵列。



在输送的电力方面可能不稳定。

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择，但并不能向所有地方稳定供电。一种解决
方案是将不需要立即使用的多余电能收集起来。
有多种技术可用来储存多余电能，包括电动汽
车所采用的锂离子电池阵列。

电池的需要有限，因为将其用于储能成本过于高昂。替代电池类型正在开
用于有效储存多余电能。钠离子电池的生产成本更低，因为使用的原材
在，应该可以实现商业化

常见主题：锂矿储量有限，获取政府许可困难，工艺不环保且成本高……

（机会） 多元化发展，进入钠矿开采及钠离子电池制造等领域。

Porrand Daily

Careers - geologists a



Continuing demand for natural resources means that demand for geologists frequently exceeds supply. An experienced geologist can expect to be well paid.

A career in geology can be challenging. There are many techniques for studying what lies beneath the earth's surface, but it is rarely possible to be 100% certain of what is there. A geologist may be confident of the presence of, say, iron ore, but the only way to be certain is to sink a test drill to collect samples. Iron ore may be present, but the ore could be low grade or the deposit may be insufficient to justify mining. There is also a possibility that the earth's structure in that area is not sufficiently stable to mine safely.

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(Risk) Competitors or other industries may grab our geologist and scientists; ensure good HR practices to recruit and retain them.

(Opportunity) Invest in new studies and techniques to find new locations that have high purity lithium (and other minerals), in stable earth structure.

(Opportunity) Work with university and research partners to support geologists and scientists who will be Rotomyne's next generation staff.

招贤纳士——需要大量地质学



地质学是
要从事岩
动。他们

我们对自
才。经验

从事地质学的工作颇具挑战性。尽管有很多技术可以用来探究地表下所蕴藏的秘密，但却无法100%确定到底是什么。地质学家可能确信存在铁矿石，但唯一确定的方法是用测试钻来收集样本。虽然可能存在铁矿石，但矿石品级可能很低，或者储量可能不足以证明具备开采的合理性。还有一种可能是，该地区的地壳结构不够稳定，无法安全开采。

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（风险）竞争对手或其他行业可能会挖走我们的地质学家和科学家；需要确保良好的HR实践，以吸引和留住他们。

（机会）投资新研究和技术，寻找具有高纯度锂（及其他矿物）的新地点，并且地质结构稳定。

（机会）与大学和研究合作伙伴合作，支持培养地质学家和科学家，他们将成为罗托矿业的下一代员工。

Exercise 6

What cyber risks does Rotomyne face?



练习6

罗托矿业面临哪些网络风险？

Types of cyber risk

Risk	Description	Controls
Strategic IT risks	• Failure of IT/IS to align with the business and competitive needs of the organisation	• Appointment of Chief Information Officer (CIO) • Develop information strategy • Maintain senior management awareness of development and potential of IT/IS
Economic risks	• Expenditure on IT/IS excessive and outweighs the business benefits • Failed IT/IS implementations	• Decentralise IT/IS expenditure to business managers • Base investment on clear deliverables and review progress periodically • Utilise successful project management (e.g. PRINCE2) and systems development methodologies • Outsource projects and operations to transfer financial risks
Availability risks	• Inability to access IT/IS systems needed for business due to technical failures • Overload of systems • Malicious Denial of Service attacks (DoS)	• Ensure testing of all hardware and software before introduction • Planned maintenance and service outages • Ensure physical security of systems • Back up sites and disaster recovery procedures • Outsource to third party provider • Load testing to assess capacity • Intrusion Prevention Systems (IPS) to automatically monitor messages to check they are legitimate, or to watch for anomalous traffic flows

网络风险类型

IT风险类型

风险	描述	控制措施
战略IT风险	IT/IS未能契合组织的业务和竞争需求	- 任命首席信息官（CIO） - 制定信息战略 - 让管理高层对IT/IS的发展和潜力保持认知度
经济风险	- IT/IS开支过高，超过商业效益 - IT/IS实施失败	- 将IT/IS开支分散到业务经理中 - 投资基于明确的可交付成果，并且定期评估进度 - 运用成功的项目管理（如PRINCE2）和系统开发方法 - 外包项目和运营，从而转移财务风险
可获得性风险	- 由于技术故障，无法访问业务所需的IT/IS系统 - 系统过载 - 恶意拒绝服务攻击（DoS）	- 确保在上线前测试所有硬件和软件 - 预先计划的维护和服务中断 - 确保系统的实体安全 - 备份站点和灾难复原程序 - 向第三方提供商外包 - 执行载荷测试，评估能力 - 入侵预防系统（IPS）自动监控讯息，检查是否合法，或观察异常流量

Types of cyber risk

Risk	Description	Controls
Security risks	• Loss of data confidentiality • Loss of data integrity due to manipulation/hacking • Loss of processing due to modification or destruction of systems	• <i>Protection approaches</i>: access identification and authentication, authorisation levels, data encryption and certificates, lock-down of personal device functionality, screening of staff and contractors • <i>Detection approaches</i>: event monitoring, intrusion detection, threat monitoring and threat intelligence (monitor the experiences of the IT community), user reports • <i>Response approaches</i>: Computer Incident Response Team (CIRT) to reduce losses, restore operations, manage crisis communications, provide decision support, and support investigations
Compliance risks	• Failure to follow laws and regulations	• Close monitoring of legal and regulatory environment • Systems audits and compliance tests • Use of Computerised-Assisted Audit Tools and Techniques (CAATTs)

网络风险类型

IT风险类型

风险	描述	控制措施
安全风险	- 失去数据保密性 - 因操纵/黑客而失去数据完整性 - 由于系统修改或破坏而失去处理	<i>保护方法</i>：访问身份识别与验证、授权级别、数据加密与证书、个人设备功能性锁定、人员和承包商筛选 <i>检测方法</i>：事件监督、入侵检测、威胁监督与威胁情报（监督IT社区的经验）、用户报告 <i>响应方法</i>：计算机事件响应小组（CIRT）负责减少损失，恢复运作，管理危机沟通，提供决策支持，支持调查
合规性风险	未能遵守法律法规	- 密切监督法律和监管环境 - 系统稽查与合规性测试 - 使用计算机辅助稽查工具与方法（CAATT）

Types of cyber risk

Risk	Description	Controls
Performance risks	• Reduced productivity/profits due to slow access to IT/IS by staff and customers	• Appropriate supplier/vendor selection process • Clear Service Level Agreements with in-house or outsourced IT/IS providers • Uniform operating procedures across systems • Steering groups to report problems and concerns • Perfective systems maintenance and development • Screening/prioritisation of user (e.g. key customer identification) • Adopt distributed processing (e.g. Cloud processing)

网络风险的类型

风险	描述	控制措施
性能风险	因员工和客户访问IT/IS速度缓慢而导致生产力/利润减少	- 适当的供应商选择流程 - 与内部或外包IT/IS提供商签订明确的服务水平协议 - 所有系统采用统一的操作程序 - 指导小组报告问题和关注点 - 完善性系统维护与开发 - 用户筛选/优先次序（如关键用户识别） - 采用分布式处理（如云端处理）

Governance and Risks at Rotomyne

罗托矿业治理与风险

Pages 9 - 11

Rotomyne's Board of Directors

- 5 Executive Directors and 4 Non-Executive Directors
 - All NEDs appointed since 2020
 - Executive Directors promoted within and appointed externally.
 - *Pre-seen does not provide indication but be aware of the possible risk of 'agency problem' – directors act in their best interest at the expense of the company/other stakeholders.*
 - *Even not tested, consider governance issues/ethics and comment in your recommendations.*
 - *You may refer to the UK Code of Governance.*
-

第9 – 11页

罗托矿业的董事会

- 5名执行董事和4名非执行董事
 - 所有非执行董事自2020年以来被任命
 - 执行董事有内部晋升和外部任命的情况。
 - 尽管预览材料未明确指出，但要注意可能存在的“代理问题”风险——董事可能会为自身利益行事，而损害公司或其他利益相关者的利益。
 - 即使不涉及考试内容，也可以考虑公司治理问题/伦理问题，并在建议中予以评论。
 - 你可以参考《英国公司治理守则》。
-

Pages 9 - 11

Rotomyne's Board of Directors

- (Strength) All Executive Directors are suitably qualified, skilled and experienced, and were promoted to the Board in the past couple of years – good performance rewarded.
 - (Risk) Most of the Eds and NEDs have had or have professional connection with customers, partners, suppliers, etc.
-

第 9 - 11

罗托矿业董事会

- （优势） 所有执行董事都具备适当的资格、技能和经验，并在过去几年中晋升为董事会成员——优秀的表现得到了回报。
 - （风险） 大多数执行董事和非执行董事曾经或目前与客户、合作伙伴、供应商等有过或仍有专业联系。
-

Exercise 7

Does Rotomyne practice good corporate governance?



练习 7

罗托矿业是否实行良好的公司治理？

Corporate Governance

- Responsibility of Board of Directors
 - appointed by shareholders to run the business
 - responsible to shareholders for how they run the business
 - 1. Must ensure effective systems of control
 - financial controls (transactions systems and budgetary control)
 - adequate management in each position
 - controls to reduce risks
 - 2. Provide direction and strategy for the business
 - Issues
 - Remuneration of senior management (pay should be related to performance and set independently by a Remuneration Committee)
 - Ensuring Board selection (should be independent process such as Nomination Committee)
 - Effective controls are in place (Audit Committees should oversee this and Board should report on its management of risk)
 - Stakeholder representation: encouragement of participation of smaller investors and methods of responding to staff concerns
 - Effective corporate board
 - Balance of Executive Directors (do) and Non-Executive Directors (advise and monitor)
 - Division of responsibility between Chairman (manages board) and CEO (manages the firm)
 - Open participation and discussion and 'strong independent element'
-

公司治理

董事会的职责

- 由股东任命，负责经营企业
- 对经营企业的状况对股东负责
 1. 必须确保有效的控制制度
 - 财务控制（交易体系和预算控制）
 - 对每个职位的充分管理
 - 降低风险的控制手段
 2. 为企业提供方向与战略

问题

- 管理高层的薪酬（薪酬应与绩效挂钩，并由薪酬委员会独立设定）
- 确保董事会成员选择（应通过独立流程，如提名委员会）
- 有效的监控（由审计委员会监督，并由董事会报告其风险管理）
- 代表利益相关方：鼓励中小投资者参与，回应员工关注问题的方法

有效的董事会

- 执行董事（行动）和非执行董事（建议与监督）之间的平衡
 - 非执行董事应保持独立性：即除了担任非执行董事的薪酬之外，与企业不存在任何商业或财务关系
 - 董事会主席（负责管理董事会）和CEO（负责公司经营管理）职责分离
 - 坦诚布公的参与与交流，“坚持独立性”
-

Rotomyne's Board Responsibilities

Shaista Shameem Chief Executive Officer			
Andrey Prokhorov Production Director	Cecilia Battistelli Human Resources Director	Martin Jacobs Chief Finance Officer (CFO)	Onwaba Makanjana Marketing Director
• Mine operations • Lithium extraction and conversion plants • Environmental protection	• Employee recruitment • Employee remuneration • Employee training • Health and safety	• Financial reporting • Management accounting • Treasury	• Sales and customer relations • Distribution channels

罗托矿业的董事会职责

沙斯塔·沙米姆 Shaista Shameem 首席执行官			
安德烈·普罗霍罗夫 Andrey Prokhorov 生产总监	塞西莉亚·巴蒂斯特利 Cecilia Battistelli 人力资源总监	马丁·雅各布斯 Martin Jacobs 首席财务官 (CFO)	奥恩瓦·马坎贾纳 Onwaba Makanjana 市场总监
• 矿场运营 • 锂提取和转化工厂 • 环境保护	• 员工招聘 • 员工薪酬 • 员工培训 • 健康与安全	• 财务报告 • 管理会计 • 财务	• 销售及客户关系 • 分销渠道

Rotomyne's Board Responsibilities (continued)

	Board committees			
	Audit	Risk	Remuneration	Nomination
Professor Iresh Jayawardena Non-Executive Chair	◆	◆		◆
Pierre Forcier Senior Independent Director	◆		◆	◆
Caroline Nguyen Ngoc Independent Non-Executive Director	◆	◆	◆	
Professor Lemi Baruh Independent Non-Executive Director		◆	◆	◆

罗托矿业的董事会职责(继续)

	董事委员会			
	审计	风险	薪酬	提名
艾瑞什·贾亚瓦德纳 (Iresh Jayawardena)教授 非执行主席	◆	◆		◆
皮尔·福瑟(Pierre Forcier) 高级独立董事	◆		◆	◆
卡罗琳·阮玉(Caroline Nguyen Ngoc) 独立非执行董事	◆	◆	◆	
莱米·巴鲁(Lemi Baruh)教授 独立非执行董事		◆	◆	◆

Exercise 8

1. What are your thoughts about Rotomyne appointing an Executive Director to oversee Legal, Risk and Compliance?
 2. What action plan you propose for the appointment of the new Executive Director?
-

练习 8

1. 你对罗托矿业任命一位执行董事负责法律、风险和合规有何看法？
 2. 你为新执行董事的任命提出的行动计划是什么？
-

Rotomyne's Principal Risk #1

- Market prices of lithium compounds are volatile, due to derived demand and supply.
 - Rotomyne maintains a close relationship with customers and manages production levels is responsive to market changes.
 - Futures and options to protect sales.
 - Be proactive and ready for 'what is coming' – supply chain management is critical.
-

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罗托矿业的主要风险 #1

- 锂化合物的市场价格波动较大，源于需求和供应的联动影响。
 - 罗托矿业与客户保持紧密合作，并通过调整生产水平灵活应对市场变化。
 - 通过使用期货和期权来保护销售。
 - 提前做好准备、预判市场趋势——供应链管理至关重要。
-

Rotomyne's Principal Risk #2

- Future production is constrained due to non availability of mines
 - cannot expand production by acquiring mines with license.
 - Rotomyne employs experienced and highly qualified professionals to monitor existing mines (supply and environmental impact) and look for potential acquisitions.
 - HR management is critical, especially staff motivation/training.
 - Investment in R&D is critical – DLE, lithium in salt water, Sulphur & Sodium mines, etc.
-

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罗托矿业的主要风险 #2

- 未来生产受限于矿山资源的短缺——无法通过收购持证矿山来扩大产能。
 - 罗托矿业聘用经验丰富且高度专业的人才，负责监控现有矿山的供应情况及环境影响，同时积极寻找潜在的收购机会。
 - 人力资源管理至关重要，尤其是在员工激励和培训方面。
 - 研发投入至关重要——重点领域包括直接锂提取（DLE）、从盐水中提取锂、以及硫矿和钠矿的开发等。
-

Rotomyne's Principal Risk #3 and #5

- Lithium mining, manufacture and sale are controlled by environmental regulations that can constrain production and attract penalties.
 - Rotomyne's mining operations are in 2 (out of 6) countries that suffer political instability.
 - Rotomyne complies or exceeds local regulations by adopting to international standards and maintains close relations with governments.
 - Government and government agencies are key players.
 - Transfer risks through local partnerships and insurance?
-

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罗托矿业的主要风险 #3和#5

- 锂矿开采、生产和销售受到严格的环保法规约束，这可能限制产能并带来罚款风险。
 - 罗托矿业在6个国家运营，其中有2个国家面临政治不稳定。
 - 罗托矿业通过采纳国际标准，遵守并超越当地法规，并与政府保持紧密关系。
 - 政府及相关机构是关键利益方。
 - 是否可以通过与当地合作伙伴合作以及购买保险来转移风险？
-

Rotomyne's Principal Risk #4

- Exchange rate fluctuations affect costs and revenues as Rotomyne mines in 6 countries and sells in 90 countries.
 - Rotomyne's internal treasury function monitors FX risks and manages them.
 - Hedging techniques: lead & lag, netting, options & futures, transactions using P\$, etc.
-

罗托矿业的主要风险 #4

- 汇率波动影响着罗托矿业的成本和收入，因其在6个国家开采矿产，并在90个国家进行销售。
 - 罗托矿业的内部财资部门负责监控和管理外汇风险。
 - 常用的对冲手段包括：提前/延迟付款、净额结算、期权与期货、以及以P\$结算交易等。
-

Rotomyne's Principal Risk #6

- Mining operations and processing involve risk of industrial accidents, causing injury (death) to employees and environmental damage.
 - Rotomyne complies with all regulations and maintain close relations with regulators and maintains transparency.
 - More training and investment for automation.
 - Contingency planning – floods, earthquake, industrial actions, etc.
-

罗托矿业的主要风险 #6

- 采矿作业和加工过程中存在工业事故风险，可能导致员工受伤甚至死亡，以及环境破坏。
 - 罗托矿业严格遵守所有法规，保持与监管机构的紧密关系，并确保运营透明。
 - 加大对员工培训和自动化的投资。
 - 制定应急预案——应对洪水、地震、罢工等突发事件。
-

Exercise 9

What additional risks would Rotomyne face with its present mining, processing and selling business? How these risks could be mitigated?

If Rotomyne were to produce a risk register, what additional information should be included?

练习 9

罗托矿业在当前的采矿、加工和销售业务中可能面临哪些额外风险？
这些风险可以如何缓解？

如果罗托矿业要制作一份风险登记表，还应包含哪些额外信息？

Terminology of Risk

‘the potential that future outcomes will be different from what was anticipated or planned’

- Risk = Impact x Probability
 - *Upside risk*: things turn out more favorably than was anticipated or planned
 - *Downside risk*: things turn out worse than was anticipated or planned
 - *Risk*: something that can be anticipated and quantified and which can be pre-empted by management action
 - *Uncertainty*: a situation where lack of data or understanding means that it is not possible to make any reliable assessment of its impact or probability
-

风险术语

“未来结果将不同于预期或计划的可能性”

- 风险 = 影响 × 概率
 - 上行风险：情况比预期或计划更加有利
 - 下行风险：情况比预期或计划更加不利

 - 风险：可以预期和量化，并且可以通过管理行动加以规避
 - 不确定性：缺乏数据或了解意味着无法对其影响或概率做出可靠评估的情况
-

Identifying risks

Long term and large

Arising from loss of managerial control

Immediate impact on business

Classifications of Risk	Strategic	Managerial	Operational
Fire, explosion injury			
Profit interest foreign exchange			
Arising from legal process			
Failure to report or operate regulations			

Breakdown in systems

Failure to compete in the market

Loss of face and integrity

辨别风险

长期，大型

因失去管理控制权而产生

对企业的直接影响

风险分类	战略风险	管理风险	营运风险
火灾，爆炸 受伤	危险		
	营运风险		
利润、利息、 外汇	财务风险		
	商业风险		
来自法律流 程	诉讼风险		
	声誉风险		
未能报告或 遵守法规	合规风险		

系统崩溃

无法开展
市场竞争

失去面子
或诚信

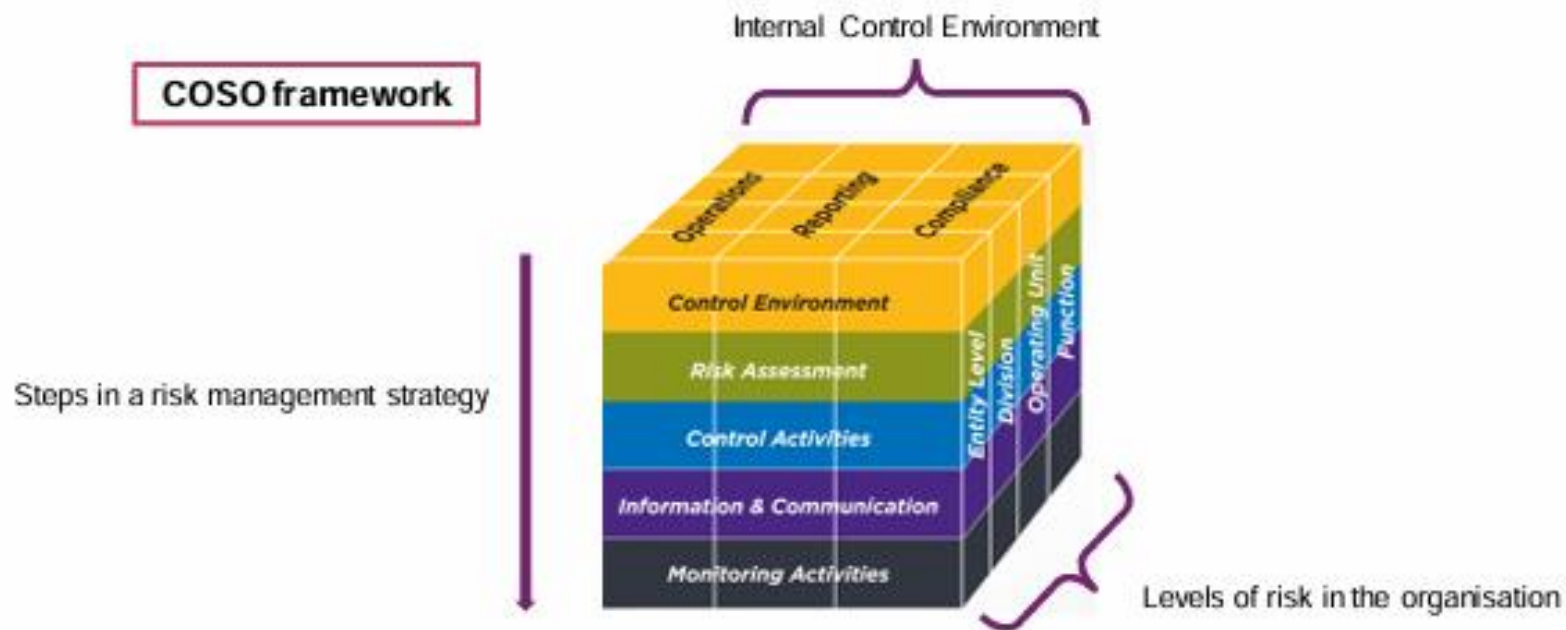
Economic and Political risk

Type of risk	Examples
Economic	• Changes in the health and growth of the economy • Volatility of exchange rates and interest rates • Changes in inflation and buyer confidence/income • Taxation changes
Political	• Unfavourable legislation • Regulation or compulsory acquisition • Corruption/bribery • Instability of government • Trade wars and sanctions

经济和政治风险

风险的类型	示例
经济	• 经济健康和增长的变化 • 汇率和利率的波动 • 通货膨胀和买方信心/收入的变化 • 税收的变化
政治	• 不利的立法 • 监管或强制收购 • 腐败和贿赂 • 政府的不稳定性 • 贸易战和制裁

COSO framework



COSO框架

Coso框架

风险管理战略的步骤



Financial Analysis of Rotomyne

罗托矿业财务分析

Rotomyne Group

Consolidated statement of profit or loss for the year ended 30 September

		2024	2023
		P\$ million	P\$ million
Revenue	- 5%	7,815	8,203
Operating costs	5%	(5,074)	(4,841)
Operating profit	- 18%	2,741	3,362
Finance costs	0%	(500)	(500)
		2,241	2,862
Tax expense	- 22%	(359)	(458)
Profit for the year	- 22%	1,882	2,404

Lithdig Group

Consolidated statement of profit or loss for the year ended 30 September

		2024	2023
		P\$ million	P\$ million
Revenue	- 5%	9,691	10,254
Operating costs	2%	(6,237)	(6,119)
Operating profit	- 16%	3,454	4,135
Finance costs	0%	(300)	(300)
		3,154	3,835
Tax expense	- 18%	(505)	(614)
Profit for the year	- 18%	2,649	3,221

Large and powerful competitor – 10 mines in 7 countries.

We have only figures for 2023 and 2024 – cannot comment on trend/market growth.

Overall decline due to energy costs and commodity prices (external factors)

Operating profit margins: Rotomyne 35% (41% in 2023) and 36% (40% in 2023)

PAT margins: Rotomyne 24% (29% in 2023) and 27% (31% in 2023)

(Strength) Rotomyne's good profit margins despite revenue drop and cost increase, and provide confidence of stability for potential investors, shareholders, customers, suppliers, etc.

(Weakness/threat) More debt than competitor (no new debt than 2023), revenue drop, costs increase

罗托矿业集团
年度合并损益表
截至 9 月 30 日

		2024	2023
		百万 P\$	百万 P\$
收入	- 5%	7,815	8,203
营运成本	5%	(5,074)	(4,841)
营业利润	- 18%	2,741	3,362
财务成本	0%	(500)	(500)
		2,241	2,862
税费	- 22%	(359)	(458)
年利润	- 22%	1,882	2,404
	- 22%		

利恩迪格集团
年度合并损益表
截至 9 月 30 日

		2024	2023
		百万 P\$	百万 P\$
收入	- 5%	9,691	10,254
营运成本	2%	(6,237)	(6,119)
营业利润	- 16%	3,454	4,135
财务成本	0%	(300)	(300)
		3,154	3,835
税费	- 18%	(505)	(614)
年利润	- 18%	2,649	3,221
	- 18%		

大型且实力强劲的竞争对手——在7个国家拥有10座矿山。我们只有2023年和2024年的数据，无法对趋势或市场增长作出长期判断。整体下滑是由于能源成本上升和大宗商品价格波动等外部因素影响。营业利润率：罗托矿业为35%（2023年为41%），对比竞争对手的36%（2023年为40%）。净利润率：罗托矿业为24%（2023年为29%），对比竞争对手的27%（2023年为31%）。

（优势） 尽管收入下滑和成本上升，罗托矿业仍保持良好的利润率，为潜在投资者、股东、客户和供应商提供了稳定信心。（劣势/威胁） 负债高于竞争对手（2023年后无新增债务），收入下降，成本增加。

Rotomyne Group**Consolidated statement of changes in equity
for the year ended 30 September 2024**

	Share capital	Retained earnings	Currency reserve	Total
	P\$ million	P\$ million	P\$ million	P\$ million
Opening balance	3,000	2,054	(217)	4,837
Profit for year		1,882		1,882
Dividend	89%	(1,684)		(1,684)
Loss on translation			(221)	(221)
Closing balance	3,000	2,252	(438)	4,814

Lithdig Group**Consolidated statement of changes in equity
for the year ended 30 September 2024**

	Share capital	Retained earnings	Currency reserve	Total
	P\$ million	P\$ million	P\$ million	P\$ million
Opening balance	3,500	4,614	(595)	7,519
Profit for year		2,649		2,649
Dividend	49%	(1,313)		(1,313)
Loss on translation			(27)	(27)
Closing balance	3,500	5,950	(622)	8,828

Huge dividend payout (89%) compared with (49% - more funds for investments).
Rotomyne's has no intention of growth in the immediate future or reliance financing it ploughing-back profit.

Currency risk not managed properly (is in Porrland and its currency loss is very less though done more business than Rotomyne)

(Strength) Rotomyne's expansions could be financed internally (if equally profitable next year), ploughing-back profit (P\$ 1.8 billion) – shareholders must be convinced that it's worth taking less dividend now for more dividend or capital growth in the future.

No indication about who are the shareholders.

罗托矿业集团
年度合并权益变更表
截至 2024 年 9 月 30 日

	股本	留存收益	货币储备	总计
	百万 P\$	百万 P\$	百万 P\$	百万 P\$
期初余额	3,000	2,054	(217)	4,837
年利润		1,882		1,882
股利	89%	(1,684)		(1,684)
折算损失			(221)	(221)
期末余额	3,000	2,252	(438)	4,814

利恩迪格集团
年度合并权益变更表
截至 2024 年 9 月 30 日

	股本	留存收益	货币储备	总计
	百万 P\$	百万 P\$	百万 P\$	百万 P\$
期初余额	3,500	4,614	(595)	7,519
年利润		2,649		2,649
股利	49%	(1,313)		(1,313)
折算损失			(27)	(27)
期末余额	3,500	5,950	(622)	8,828

巨额股息支付率（89%），相比之下，竞争对手为49%——竞争对手保留了更多资金用于投资。罗托矿业近期无意进行扩张，也不打算通过利润再投资来进行融资。汇率风险管理不当（竞争对手位于波兰德，尽管业务量超过罗托矿业，但汇率损失较小）。

（优势） 如果明年同样盈利，罗托矿业可以通过内部融资支持扩展（将P\$18亿利润再投资）。需要说服股东暂时接受较低的股息，以期未来获得更高的股息或资本增值。目前没有关于股东身份的明确信息。

Rotomyne Group
Consolidated statement of financial position
as at 30 September

	2024	2023
	P\$ million	P\$ million
Assets		
Non-current assets		
Property, plant and equipment	7,270	7,344
Goodwill	1,814	1,814
Other intangible assets	+11% 326	294
	<u>9,410</u>	<u>9,452</u>
Current assets		
Inventory	845	866
Trade receivables	765	879
Bank	857	852
	<u>2,467</u>	<u>2,597</u>
Total assets	<u>11,877</u>	<u>12,049</u>
Equity		
Share capital	3,000	3,000
Currency reserve	(438)	(217)
Retained earnings	2,252	2,054
	<u>4,814</u>	<u>4,837</u>
Liabilities		
Non-current liabilities		
Borrowings	5,000	5,000
Current liabilities		
Trade payables	1,700	1,755
Tax liability	363	457
	<u>2,063</u>	<u>2,212</u>
Total equity and liabilities	<u>11,877</u>	<u>12,049</u>

seems to have done acquisitions in the past (20% increase in PPE).

11% increase in intangible assets - intellectual property rights, technology advancements etc.

Trade payables are almost x2 as cash in the bank.

We hold less inventory (-2%) compared with Lithididg (+43%) than 2023.

Common Exam Mistake: Retained Earnings P\$ 2,252 is the cumulative total and all of it is not available for investment purposes!

Lithdig Group
Consolidated statement of financial position
as at 30 September

	2024	2023
	P\$ million	P\$ million
Assets		
Non-current assets		
Property, plant and equipment	8,797	7,344
Goodwill	2,045	2,045
Other intangible assets	418	392
	<u>11,260</u>	<u>9,781</u>
Current assets		
Inventory	1,530	1,074
Trade receivables	786	1,104
Bank	744	1,044
	<u>3,060</u>	<u>3,222</u>
Total assets	<u>14,320</u>	<u>13,003</u>
Equity		
Share capital	3,500	3,500
Currency reserve	(622)	(595)
Retained earnings	5,950	4,614
	<u>8,828</u>	<u>7,519</u>
Liabilities		
Non-current liabilities		
Borrowings	3,000	3,000
Current liabilities		
Trade payables	1,984	1,872
Tax liability	508	612
	<u>2,492</u>	<u>2,484</u>
Total equity and liabilities	<u>14,320</u>	<u>13,003</u>

罗托矿业集团 合并财务状况表 截至 9 月 30 日		
	2024 百万 P\$	2023 百万 P\$
资产		
非流动资产		
物业、厂房和设备	7,270	7,344
商誉	1,814	1,814
其他无形资产	+11% 326	294
	9,410	9,452
流动资产		
库存	845	866
应收账款	765	879
银行	857	852
	2,467	2,597
总资产	11,877	12,049
权益		
股本	3,000	3,000
货币储备	(438)	(217)
留存收益	2,252	2,054
	4,814	4,837
负债		
非流动负债		
借贷	5,000	5,000
流动负债		
应付账款	1,700	1,755
纳税义务	363	457
	2,063	2,212
总权益和负债	11,877	12,049

公司似乎在过去进行过收购（固定资产增长了20%）。
无形资产增长了11%——主要是知识产权、技术提升等。
应付账款几乎是银行存款的两倍。我们持有的库存相比2023年减少了2%，而Lithididg则增长了43%。
常见的考试错误：留存收益P\$ 2,252是累计数额，并不是全部都可以用于投资！

利恩迪格集团 合并财务状况表 截至 9 月 30 日		
	2024 百万 P\$	2023 百万 P\$
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非流动资产		
物业、厂房和设备	8,797	7,344
商誉	2,045	2,045
其他无形资产	418	392
	11,260	9,781
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	8,828	7,519
负债		
非流动负债		
借贷	3,000	3,000
流动负债		
应付账款	1,984	1,872
纳税义务	508	612
	2,492	2,484
总权益和负债	14,320	13,003

Rotomyne Group
Consolidated statement of financial position
as at 30 September

	2024	2023
	P\$ million	P\$ million
Assets		
Non-current assets		
Property, plant and equipment	7,270	7,344
Goodwill	1,814	1,814
Other intangible assets	326	294
	9,410	9,452
Current assets		
Inventory	845	866
Trade receivables	765	879
Bank	857	852
	2,467	2,597
Total assets	11,877	12,049
Equity		
Share capital	3,000	3,000
Currency reserve	(438)	(217)
Retained earnings	2,252	2,054
	4,814	4,837
Liabilities		
Non-current liabilities		
Borrowings	5,000	5,000
Current liabilities		
Trade payables	1,700	1,755
Tax liability	363	457
	2,063	2,212
Total equity and liabilities	11,877	12,049

Working Capital Ratios:

Receivable days = 36 (Lithidig 30)

Payable days = 122 (Lithidig 116.8)

Working capital cycle = - 47 (Lithidig - 29)

Liquidity Ratios:

Current ratio = 1.20 (Lithidig 1.23)

Quick ratio = 0.79 (Lithidig 0.61)

Interest cover = 5.48 (Lithidig 11.51)

(Interest rate = 10% for both companies)

Gearing Ratios:

LT Debt/Equity = 104% (Lithidig 34%)

LT Debt/LT Debt + Equity = 51% (Lithidig 25%)

(Weakness) Rotomyne is not very liquid

(Weakness) If Rotomyne requires additional funding for expansion/investment, long term borrowing will be challenging due to high gearing (and declining profit, lower interest cover than Lithidig). Further, 89% dividend means and high debt: where funds will come for investment?

Lithidig Group
Consolidated statement of financial position
as at 30 September

	2024	2023
	P\$ million	P\$ million
Assets		
Non-current assets		
Property, plant and equipment	8,797	7,344
Goodwill	2,045	2,045
Other intangible assets	418	392
	11,260	9,781
Current assets		
Inventory	1,530	1,074
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Total assets	14,320	13,003
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Retained earnings	5,950	4,614
	8,828	7,519
Liabilities		
Non-current liabilities		
Borrowings	3,000	3,000
Current liabilities		
Trade payables	1,984	1,872
Tax liability	508	612
	2,492	2,484
Total equity and liabilities	14,320	13,003

罗托矿业集团 合并财务状况表 截至 9 月 30 日		
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	4,814	4,837
负债		
非流动负债		
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应付账款	1,700	1,755
纳税义务	363	457
	2,063	2,212
总权益和负债	11,877	12,049

营运资本比率：：

应收账款天数 = 36天（利恩迪格 30天）应付账款天数 = 122天（利恩迪格 116.8天）营运资本周期 = -47天（利恩迪格 -29天）

流动性比率：：

流动比率 = 1.20（利恩迪格 1.23）速动比率 = 0.79（利恩迪格 0.61）利息覆盖率 = 5.48（利恩迪格 11.51）（两家公司利率均为10%）

杠杆比率：：

长期负债/股本比率 = 104%（利恩迪格 34%）长期负债/（长期负债 + 股本）比率 = 51%（利恩迪格 25%）

（劣势） 罗托矿业的流动性较差。

（劣势） 如果罗托矿业需要额外资金用于扩展或投资，考虑到其较高的负债率、利润下降以及低于利恩迪格的利息覆盖率，长期借贷将面临挑战。此外，89%的股息支付率和高负债水平意味着公司投资资金的来源将非常有限。

利恩迪格集团 合并财务状况表 截至 9 月 30 日		
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应付账款	1,984	1,872
纳税义务	508	612
	2,492	2,484
总权益和负债	14,320	13,003

Exercise 10

Is Rotomyne's management delivering a satisfactory financial performance for its shareholders?

练习 10

罗托矿业的管理层是否为股东带来了令人满意的财务业绩？

Exercise 11

1. What factors may have caused the changes in share prices shown on page 17?
 2. What is meant by a beta coefficient?
 3. What factors determine a firm's beta coefficient?
-

练习11

1. 导致第17页股价变化的因素可能包括哪些？
 2. 什么是Beta系数？
 3. 哪些因素决定了公司的Beta系数？
-

- We'll assume 1.63 is Rotomyne's equity (geared) beta (reflects both the business risk of being a mining company and the financial risk associated with Rotomyne's current gearing level:
 - book values of equity and debt are approximately equal to market values
 - the tax rate is approximately 16% (in line with Rotomyne's current tax rate)
 - debt finance is approximately risk free (so debt beta is zero).

$$\beta_{\text{ungeared}} = \left[\frac{V_E}{V_E + V_D[1 - T]} \right] \beta_{\text{geared}}$$

$$\beta_{\text{ungeared}} = \left[\frac{4,814}{4,814 + 5,000[1 - 0.16]} \right] 1.63$$

Lithium mining industry has a lower-than-average level of systematic business risk (the ungeared beta is well below 1).

Rotomyne has a high level of gearing, its equity beta is much higher. Rotomyne faces a higher overall level of systematic risk than the average company in the market.

- 我们假设罗托矿业的权益（杠杆）Beta系数为1.63（反映了作为采矿公司的业务风险以及与当前杠杆水平相关的财务风险）：
 - 股权和债务的账面价值大致等于市场价值
 - 税率约为16%（与罗托矿业当前税率一致）
 - 债务融资几乎无风险（因此债务Beta为零）

$$\beta_{\text{ungeared}} = \left[\frac{V_E}{V_E + V_D[1 - T]} \right] \beta_{\text{geared}}$$

$$\beta_{\text{ungeared}} = \left[\frac{4,814}{4,814 + 5,000[1 - 0.16]} \right] 1.63$$

锂矿行业的系统性业务风险低于平均水平（无杠杆的Beta值远低于1）。

罗托矿业的杠杆水平较高，因此其权益Beta值远高于行业平均水平。罗托矿业面临的整体系统性风险高于市场上的平均公司。

Factors influencing share price

Factor	Description
Future earnings (forecast free cash flows)	• Market value assumed to be related to PV of future earnings (free cash flows) • Past earnings may be used as a basis or indicator for this forecast
Risks of business (impacts the rate of discount applied to cash flows)	• Potential for unanticipated increases (hope) or falls (problems) in earnings or value • Capital structure of the business • Exposure to external events that impact on its business model
Dividends (impacts when cash will be received by investor)	• Share price will rise when shares are 'cum div' • Changes in dividend policy tend to move share price in same direction (contrary to dividend irrelevance hypothesis)
Rumours and events affecting the business	• Changes of management or strategy • Major client wins or product discoveries • Speculation about acquisition activity • Potential for favourable or adverse regulations coming into force
Industry growth and outlook	• Growth rate of the industry as a whole • External impacts affecting the industry (costs, exchange rates, competition)
Investment market sentiment	• Poor economic outlook will reduce all share prices • High market liquidity (cash) and low interest rates will boost all share prices

影响股价的因素

因素	说明
未来盈利 (预测自由现金流量)	• 市场价值被假设为与未来盈利（自由现金流）的现值相关联 • 以往盈利可用作这一预测的基础或指标
商业风险 (影响现金流折现率)	• 存在盈利或价值意外上涨（希望）或下跌（问题）的可能性 • 企业的资本结构 • 对影响其商业模式的外部事件的风险敞口
股利 (影响投资者何时收到现金)	• 如果股票“带股利”，则股价将上涨 • 股利政策的变动往往会造成股价朝相同方向移动（与股利不相关 假定相反）
影响企业的传闻与事件	• 管理层或企业战略变动 • 赢得大客户或产品发现 • 有关收购活动的猜测 • 有利或不利法规实施的可能性
行业增长与前景	• 行业整体的增长率 • 影响行业的外部影响（成本、汇率、竞争）
投资市场情绪	• 糟糕的经济前景将造成所有股价下跌 • 较高的市场流动性（现金）和较低的利率将造成所有股价上涨